ROYAL CANADIAN NAVY
QUALIFICATION STANDARD AND PLAN (QSP)

Naval Electronic Sensor Operator (NES Op) (00115)

Rank Qualification – Petty Officer 2nd Class (RQ-PO2)

Electronic Warfare Supervisor (EWS)

NQUAL – ALBB

MITE ID – 121756

RECORD OF AMENDMENTS

The training authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Centre Atlantic (NTDC (A)) is the designated development centre responsible for the maintenance of training. Amendments to PO standard are to be requested through NTDC (A) and approved by CNPTG. Amendments to Lesson Guidance are to be approved by CO NTDC (A).

Revision	Approval Date	Authority
2.0	27 Jun 19	NPTG
3.0	26 Jan 21	NTDC(A)

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FOREWORD

1. This training is based on the specification found at:

   a. A-P9-050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

   b. A-PD-055-002/PP-003 Canadian Forces Officer General Specification dated 23 Apr 2014;

   c. A-PD-055-002/PP-002 Non-Commissioned Member General Specification dated 09 Feb 2011;

   d. A-PD-055-002/PP-001 Maritime Surface and Sub-surface (MARS) 00207 Job-Based Occupational Specification dated 01 Mar 2015;

   e. A-PD-055-002/PP-002 Naval Electronic Sensor Operator (NES Op) 00115 Job Based Occupational Specification dated 26 Aug 2014;

   f. A-PD-055-003/PQ-001 Above Water Warfare Director (AWWD) Occupational Specialty Specification dated 11 May 2016;

   f. NAVORD 4500-00; and

   g. NPTORD 4500-22.

2. A trainee will achieve the qualification ALBB when the standards listed in all Performance Objectives have been met.

3. A trainee will achieve the following MITE Course ID 121756 when standards listed in the Performance Objectives have been met.

4. The designated TEs for this training are indicated below:

TE	Phase / Module	MITE course ID	NQual
NFS(A) / NFS (P)	All	121756	ALBB

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the NES Op RQ PO2 ALBB. This QSP is designed to assist NTDC(A) in developing, Naval Fleet Schools in conducting and evaluating, and NPTG in validating the training.

COURSE CAPACITY

2. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40.

   a. Maximum – 12;

   b. Minimum – 4; and

   c. Optimum – 8.

3. If course capacity is over optimum, a supplemental instructor is required for the entirety of PO 002 and PO 003.

COURSE DURATION

4. Course duration is calculated in accordance with (IAW) NPTORD 4500-22, Annex B. For this course, a period of instruction is 50 minutes in duration and each training day is comprised of 8 periods. These calculations are reflective of maximum load. Scheduling cells are to increase/decrease course length based on number of trainees enrolled on the course. Note that partial days are rounded up.

Course Duration Calculation		
Allocation of ATT	Periods	Days
Sum of all EOs/POs	273	34.1
Enabling Checks (ECs)	71	8.8
Performance Checks (PCs)	255	31.9
ATT SUB TOTAL:	599	74.8
Allocation of BAT, TAT, TTT and PT	Periods	Days
Base Administrative Time (BAT)	19	2.4
Training Administrative Time (TAT)	19	2.4
Training Travel Time (TTT)	0	0
Physical Fitness (PT)	51	6.3
BAT, TAT, TTT, PT SUB TOTAL:	89	10.5

(ATT+BAT+TTT+TAT+PT) 400 minutes			
	TOTAL:	688	86

REQUIRED COURSE INSTRUCTORING

5. Identify key personnel required to conduct the training.

Position	Rank	Qualifications	Qty	PO / EO
Course Officer	Lt(N)	AWWD	1	Nil
Instructor	PO2	NES Op RQ-PO2	1	PO 001-003
Instructor	PO2	NES Op RQ-PO2	1	PO 001-003
RCN Trinity	Any	Familiar with the RCN Trinity website	1	EO 001.03
CFMWC	PO1	SME in Naval Gun Fire Support	1	EO 003.07
*Supplemental Instructor	PO1/PO2	NES Op RQ-PO2	1	PO 001-003

*See course capacity para 3.

PREREQUISITES

6. To be selected for this course, the loading authority must ensure that trainees have the following prerequisites:

a. Be a qualified Fire Control Supervisor (AKCL); and

b. Hold a security clearance of at least Level 2.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

7. PLARs shall be conducted IAW NPTORD 4500-27, DAOD 5031-1 and CFITES Vol. 12.

SPECIAL INSTRUCTIONS

8. Nil.

CHAPTER 2: PERFORMANCE OBJECTIVES & LESSON GUIDANCE

PO 001 – Prepare Electronic Warfare Correspondence

1. Performance: Prepare Electronic Warfare Correspondence

2. Conditions:

   a. Given:
      (1) Command direction,
      (2) Computer message drafting software,
      (3) OPTASK/OCS/TABORD/OPSTAT messages,
      (4) References,
      (5) Briefing note template,
      (6) Mission scenarios,
      (7) Supervision, and
      (8) Open source Naval/Air Platform information,

   b. Denied: nil; and

   c. Environmental: any.

3. Standard: the Electronic Warfare Supervisor (EWS) shall prepare Electronic Warfare correspondence by:

   a. Preparing an OPTASK EW SUPP by:
      (1) Extracting EW information from OPGEN/OPORD/OPTASK and OPSTAT messages,
      (2) Interpreting Electronic Warfare orders (C00530), and
      (3) Drafting an OPTASK EW Supp,

   b. Preparing an Emission Control Plan (EMCON) (B00180)
      (1) Interpreting emission control policies (C00650),
      (2) Extracting information from Communication Plans (COMPLANS), Emission Control Plans (EMCON) (A00040/50),
      (3) Adhering to EMCON Restrictions (B00120),
      (4) Preparing EMCON Table (B00190), and
      (5) Reacting to changes in the EW environment (C00580),

   c. Evaluating Intelligence Information (B00090) by:
      (1) Operating C4I systems (F00210),
      (2) Compiling electronic intelligence reports, and
      (3) Drafting an EW briefing note,

   d. Performing Threat Analysis (C00540) by:
      (1) Identifying Maritime, Air and Land based platforms (F00180), and

(2) Preparing EW brief utilizing presentation application (C00560), and

e. Conducting EW Operational Training (C00280) by:

(1) Preparing operational training exercises (B00200) by:

(a) Planning/designing training exercises (G00050) by:

(i) Producing OCS/TABORD messages for serial planning,

(ii) Inputting training scenarios (G00230), and

(iii) Updating training scenarios (G00110),

(2) Conducting individual training events (G00020) to include:

(a) Assisting in individual training events (G00010),

(b) Assessing individual training performance (G00070), and

(c) Conducting Unit On Job Training (OJT), Ships Development Package (SDP), Pre-Information Package (PIP) (G00190), and

(3) Conducting team training events (G00090) by:

(a) Assisting in team training events (G00180), and

(b) Assessing team training performance (G00140).

4. Training Limitations: Nil.

EO 001.01 – Prepare OPTASK EW SUPP

001.01.01 – Operational Messages

Conditions		Sequence
Classified Classroom		After PO 002
Time	**Classification**	**Instructor**
1050 minutes	SECRET	NES Op PO1/PO2

Learning Objectives	Method
1) Extracting EW information from OPGEN/OPORD/OPTASK and OPSTAT messages 2) Interpret Electronic Warfare orders 3) Draft an OPTASK EW Supp	• Group Study • Floating Flip Chart • Practical

Activity Steps	
	<u>LO 1&2: Group Study (70 minutes)</u> 1) Class is to be divided into 3 groups. Each group will review the following message examples: a. OSTAT Units b. OPGEN/OPGEN SUPP c. OPTASK AAW/AWW SUPP d. OPTASK ASuW/ASuW SUPP e. OPTASK AD/AD SUPP f. OPTASK SCC/SCC SUPP g. OTASK EW/EW SUPP 2) Meanwhile, the instructor will divide the white board into 3 ‘flipcharts’, then divide each message type amongst them. <u>LO 1&2: Floating Flip Chart (40 minutes)</u> 3) Each group will be assigned to a flipchart and list information located in the titled messages. 4) Upon completion, each group will rotate to a different flipchart, review the previous group’s findings and add to them. 5) Repeat until each group has gone through each flipchart. 6) Once at their final chart, each group will assign a spokesperson to debrief the findings. 7) Instructor are encouraged to ask follow-on questions such as: a. What information is contained in each message? b. Who is responsible for drafting each message? c. Who is the approving authority and/or who are the various addressees for each message? <u>LO 3: Practical (940 minutes)</u> 8) Instructor will provide a quick refresher of message drafting software and review the OPTASK EW message format outlined in APP-11.

	9) Trainees will prepare an OPTASK EW message based on a predetermined threat Game Pack assigned. 10) Trainees will submit their message to the instructor for corrections and/or feedback.

Resources	References
1) Secret standalone workstation with access to printer 2) Message drafting software 3) CANFLTLANT website 4) Standing OPGEN 5) Standing OPTASK messages 6) NTDC(A) approved scenario	1) AIT manual 2) APP-11 3) CFCD-114

Notes to Instructor

- For best conduct of Floating Flip Chart activity refer to AIT manual (p. 23).

EC 001.01 – Prepare OPTASK EW SUPP

Scope	Conditions	Sequence
EO 001.01	Classified Classroom	After EO 001.01
Time	**Classification**	**Instructor**
400 minutes	SECRET	NES Op PO1/PO2

Assessment Steps	**Input:** Trainees will be provided with a copy of EC 001.01 checklist, be given time to review and ask clarifying questions as required. Trainees will then be provided with the following: 1) NTDC(A) approved scenario 2) OPTASK EW **Performance:** Trainees will produce 1 OPTASK EW SUPP. **Output:** The trainee must achieve 17/22 SATs to pass this EC. **Feedback:** Instructor will debrief trainees individually following completion of the EC.

Resources	References
1) OPTASK EW 2) OPGEN 3) Secret standalone workstation with access to printer 4) CANFLTLANT website	1) APP-11 2) CFCD-114

Notes to Instructor
Nil.

Limitations	Nil.
Impact of Failure	IAW NPTORD 4500-54.

Item	EC 001.01 PREPARE OPTASK EW SUPP	SAT	UNSAT	Remarks
1. OPTASK EW SUPP IAW APP-11				
Message identification - Exercise nickname - Exercise identifier		☐	☐	
Operation codeword - Operation codeword - Plan originator and number - Option nickname - Secondary option nickname		☐	☐	
Message identifier - Message text format identifier - Standard - Version - Originator - Message serial number - Reference time of publication - Qualifier - Serial number of qualifier - Message security policy - Message security classification - Message security category		☐	☐	
References - Serial letter - Communication type - Originator - Date and/or time of reference - Reference serial number - Special notation - SIC or file number		☐	☐	
Period of time - Start day-time - Stop day-time - Qualified stop time		☐	☐	
Conduct of operations - Text indicator - Instructions present (free text)		☐	☐	
Enemy electronic order of battles - Priority identifier - Emitter category		☐	☐	

- Emitter designation, nickname or spot number - Nation of emitter			
Target of interest - Priority identifier - Emitter category - Radar frequency - Pulse repetition frequency - Pulse repetition interval - Pulse width - Scan period - Nation of emitter	☐	☐	
Duty assignment details - Unit or organization - Duty code	☐	☐	
Electronic support measures tasking - Equipment designator - Frequency limits or spot number - Unit	☐	☐	
Tasking of jammers - Platform carrying equipment - Jamming control authority for unit - Jammer type to be used - Frequency limits or spot number - Jammer alert state	☐	☐	
CHAFF employment instructions - Text indicator - Instructions present (free text)	☐	☐	
Emission control plan format - Index number or letter - Equipment or tactical unit	☐	☐	
Emission control plan details - Emission control plan details - Index letter(s) - Equipment index and radiation status indicator	☐	☐	
Force electronic orders of battles - Unit name - Correction to EOB - EW equipment type	☐	☐	
Prevention of mutual interference instructions - Text indicator	☐	☐	

- Instructions present (free text)			
Satellite countermeasures instructions - Text indicator - Instructions present (free text)	☐	☐	
Satellite vulnerability details - Satellite system name or designator - Start time of satellite vulnerability - Stop time of satellite vulnerability	☐	☐	
Reporting instructions - Text indicator - Instructions present (free text)	☐	☐	
Special instructions - Text indicator - Instructions present (free text)	☐	☐	
Special information - Text indicator - Instructions present (free text)	☐	☐	
Acknowledge requirement - Acknowledge requirement indicator - Instructions for force/unit required to acknowledge	☐	☐	

Score: / 22
Trainee must achieve 17/22 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.	**Trainee** I have been debriefed on the result of this assessment.

Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

PROTECTED B (when complete)

NTDC(A) Approval – 26/01/2021
NPTG HQ Approval – 27/06/2019
Version No: 3.0
MITE Course ID: 121756

EO 001.02 – Prepare an Emission Control Plan Table

001.02.01 – Electronic Warfare Planning

Conditions		Sequence
Classified Classroom		After EO 001.01
Time	**Classification**	**Instructor**
600 minutes	SECRET	NES Op PO1/PO2

Learning Objectives	Method
1) Interpret EMCON policies 2) Extract information contained in EMCON plans 3) Adhere to EMCON restrictions 4) Prepare EMCON table 5) React to changes in the EMCON environment	• Self-Study • Table Topics • Practical • Interactive Lecture

Activity Steps

<u>LO 1&2: Self-Study (450 minutes)</u>

1) Trainees will conduct preliminary readings relating to emission control from the NTDC(A) PO 001 Required Reading List.
2) Meanwhile, the instructor will prepare EMCON related statements, questions or subjects. Use table 8-1/8-2 of ATP-01 Vol I in order to prepare Table Topic statements. Examples of topics include:
    a. Define Emission Control
    b. What factors affect the EMCON
    c. Name 3 EMCON considerations
    d. Describe the EMCON objectives

<u>LO 1&2: Table Topic (20 minutes)</u>

3) The statements are put into a bag and trainees are going to draw one in their turn. Ensure trainees do not to draw ahead of time to ensure they are focusing on their peers responses.
4) Each trainee will speak on the topic they have drawn. Once completed, remainder of the class is encouraged to share their comments on the subject. Then the bag is passed to another learner and the process is repeated until all subject have been exhausted.
5) What information is contained in each message?
    a. Who is responsible for drafting each message?
    b. Who is the approving authority and/or who are the various addressees for each message?

<u>LO 3: Practical (130 minutes)</u>

6) Trainees will prepare an EMCON table based on OPTASK EW SUPP drafted during EO 001.01.
7) Trainees will submit their message to the instructor for corrections and/or feedback.

Resources	References
1) Blank EMCON table 2) RSI table 3) Message drafting software 4) Secret standalone workstation and access to printer	1) AIT Manual 2) ATP-01 Vol 1 3) ATP-31 4) CFCD-114

Notes to Instructor

- For best conduct of Table Topic activity refer to AIT manual (~p. 26).

EC 001.02 – Prepare EMCON TABLE

Scope	Conditions	Sequence
EO 001.02	Classified Classroom	After EO 001.02
Time	**Classification**	**Instructor**
100 minutes	SECRET	NES Op PO1/PO2

Assessment Steps	
	Input: Trainees will be provided with a copy of EC 001.02 checklist, be given time to review and ask clarifying questions as required. Trainees will then be provided with the following: 1) Standing OPTASK EW 2) OPTASK EW SUPP 3) Blank EMCON table **Performance:** Trainees will produce 1 EMCON table each. **Output:** Trainee must be successful in all sections to pass this EC. The trainee must achieve: 1) Section I: 8/10 SATs 2) Section II: 1/1 SAT 3) Section III: 3/4 SATs 4) Section IV: 1/1 SAT **Feedback:** Instructor will debrief trainees individually following completion of the EC.

Resources	References
1) Blank EMCON table 2) Standing OPTASK EW 3) OPTASK EW SUPP 4) Secret standalone workstation with access to printer	1) ATP-01 Vol 1 2) ATP-31 3) CFCD-114

Notes to Instructor
Nil.

Limitations	Nil.
Impact of Failure	IAW NPTORD 4500-54.

Item	EC 001.02 PREPARE AN EMISSION CONTROL PLAN TABLE	SAT	UNSAT	Remarks
1. Section I – General EMCON Considerations Taken IAW ATP-01 Vol 1				
Reflect/support OTC/CWC aim		☐	☐	
Accounts for enemy threat		☐	☐	
Accounts for enemy capabilities		☐	☐	
Accounts for enemy tactics		☐	☐	
Cover all aspects of warfare including; - Electro-Optical - Bridge dial illumination - Cigarette glows considerations - De-icing equipment - Funnel exhaust - Signal projector - Radiated acoustic noise - Operation of auxiliary machinery - Non-essential crew activity		☐	☐	
Provide sufficient time warning for the use of pickets		☐	☐	
Allow for reaction time of sensors and weapons that have been silent		☐	☐	
Allow for ROE in use		☐	☐	
Reflect the disposition of the following; - Maritime units - Areas of operations of supporting A/C - Surface ships - Submarines		☐	☐	
Allow for the following; - Periodic system check - Periodic system maintenance		☐	☐	
2. Section II – Tactical Situation Considerations Taken IAW ATP-01 Vol 1				
Table has one specific plan for each of the following TACSIT; - Unknown - Force located and targeted - Force location known; disposition unknown - Force not located		☐	☐	
3. Section III – EMCON Objectives Met IAW ATP-01 Vol 1				
Gain maximum information from enemy		☐	☐	

Deny information to enemy	☐	☐	
Includes all emitters	☐	☐	
Easy to understand	☐	☐	
4. Section IV – Radar Status Indicator Employment IAW ATP-01 Vol 1			
RSI employed IAW ATP-01 Vol 1	☐	☐	

Score:
Section I /10
Section II /1
Section III /4
Section IV /1

Trainee must be successful in all sections to pass this EC. The trainee must achieve:

1) Section I: 8/10 SATs
2) Section II: 1/1 SAT
3) Section III: 3/4 SATs
4) Section IV: 1/1 SAT

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the trainee.		**Trainee** I have been debriefed on the result of this assessment.	
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

PROTECTED B (when complete)

NTDC(A) Approval – 26/01/2021
NPTG HQ Approval – 27/06/2019
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EO 001.03 – Evaluate Intelligence Information

001.03.01 – Operate C4I Systems

Conditions		Sequence
CSNI Lab		After EO 001.02
Time	**Classification**	**Instructor**
350 minutes	SECRET	NES Op PO1/PO2 Guest Lecturer (if available)

Learning Objectives	Method
1) Operate C4I Systems	• Interactive Lecture

Activity Steps	
	<u>LO 1: Interactive Lecture (350 minutes)</u> 1) Using the NTDC(A) Network Road Map checklist, instructor will guide trainees through the networks and all EW related sites and their content. 2) Instructor to discuss NSWAN. 3) Instructor is to discuss how the listed sites can be of use to an EWS

Resources	References
1) Secret network 2) CNSI lab 3) NTDC(A) Network Road Map checklist 4) Projector	Nil.

Notes to Instructor
• Guest lecturer from RCN Trinity.

EO 001.03 – Evaluate Intelligence Information

001.03.02 – Intelligence Exploitation

Conditions		Sequence
CSNI Lab		After EO 001.02.01
Time	**Classification**	**Instructor**
500 minutes	SECRET	NES Op PO1/PO2 Guest Lecturer (if available)

Learning Objectives	Method
1) Compile electronic intelligence reports 2) Draft an EW briefing note	• Demonstration • Practical

Activity Steps	
	<u>LO 1: Demonstration (100 minutes)</u> 1) Instructor will provide trainees with access to various Daily Intelligence sites then: a. Assist with extracting and evaluating pertinent information b. Provide guidance on how to make a proper assessment c. Explain how this information can affect mission success or daily operations <u>LO 2: Practical (400 minutes)</u> 2) Trainees will draft a briefing note based on their findings utilizing the CAF writing guide. Briefing note shall include points that would likely impact their unit and or mission with an individual recommendation to Command

Resources	References
1) Standalone workstation with access to printer 2) Briefing note template 3) Intelligence websites 4) CSNI Lab	1) CFSAL Military Writing Guide

Notes to Instructor
Nil.

EC 001.03 – Evaluate Intelligence Information

Scope	Conditions	Sequence
EO 001.03	CSNI lab	After EO 001.03
Time	**Classification**	**Instructor**
200 minutes	SECRET	NES Op PO1/PO2

Assessment Steps	
	Input: Trainees will be provided with a copy of EC 001.03 checklist, be given time to review and ask clarifying questions as required. Trainees will then be provided with a threat platform. **Performance:**; 1) Trainees will conduct topic research in the CSNI lab (100 minutes) 2) Trainees will draft an EW briefing note based on their findings on provided topic (100 minutes) EW briefing note is to concentrate on EW capabilities of threat platform and contain a recommendation on employment of own sensors to gain/deny information to the enemy. Trainees will produce 1 EW briefing note each. **Output:** The trainee must achieve 4/5 SATs to pass this EC. **Feedback:** Instructor will debrief trainees individually following completion of the EC.

Resources	References
1) CSNI lab	1) CFSAL Military Writing Guide

Notes to Instructor
• Multiple trainees can be assigned the same threat platform and research can be completed as a group, however, the EW briefing note must be drafted individually.

Limitations	Nil.
Impact of Failure	IAW NPTORD 4500-54.

Item	EC 001.03 EVALUATE INTELLIGENCE INFORMATION	SAT	UNSAT	Remarks
1. Prepare EW briefing note				
EW briefing note follows CAF writing guide briefing note template format		☐	☐	
Security classification correct		☐	☐	
Contains assessment on the following; - Threat posed by platform - Enemy tactics - Enemy sensors employment - Enemy weapons employment		☐	☐	
Contains recommendation on the following: - Own sensor employment - Own weapons employment - Tactics employment		☐	☐	
Spelling, grammar and punctuation 5 or fewer errors		☐	☐	

Score: /5
Trainee must achieve 4/5 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.	**Trainee** I have been debriefed on the result of this assessment.

Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

EO 001.04 – Perform Threat Analysis

001.04.01 – Order of Battle

Conditions		Sequence
CSNI lab		After EO 001.03
Time	**Classification**	**Instructor**
1000 minutes	SECRET	NES Op PO1/PO2

Learning Objectives	Method
1) Conduct/Participate threat analysis including the collection, analysis and dissemination of electronic data 2) Identify Maritime, Air and Land based platforms 3) Compile Operational Order of Battle 4) Prepare an EW brief using presentation application	• Collaborative Problem Solving

Activity Steps	
	<u>LO 1 to 4: Interactive Lecture (1000 minutes)</u> 1) Instructor will provide trainees with an Order of Battle then divide the class into 3 groups. a. Group 1 will be tasked to conduct research and create the Surface Threat Sheet. b. Group 2 will be tasked to conduct appropriate research and create the Air Threat Sheet. c. Group 3 will be tasked to conduct appropriate research and create the TELAR and SAM sites Threat Sheets. 2) Upon completion, each group will present their findings. 3) Instructor will provide amplifying information as required to ensure the information presented entails a full EW brief.

Resources	References
1) CSNI lab 2) Projector 3) Threat sheet template 4) TDC(A) OOB 5) TRINITY website 6) NEWC website	Nil.

Notes to Instructor
• Pick an OOB from the CA approved Game Packs.

EC 001.04 – Perform Threat Analysis

Scope	Conditions	Sequence
EO 001.04	CSNI lab	After EO 001.04
Time	**Classification**	**Instructor**
400 minutes	SECRET	NES Op PO1/PO2

Assessment Steps	
	Input: Trainees will be provided with a copy of EC 001.04 checklist, be given time to review and ask clarifying questions as required. Trainees will then be provided with a random surface and air platform from the submitted threat sheets. **Performance:**; 1) Trainees will conduct topic research and compile findings in the CSNI lab (250 minutes) 2) Trainees will present their findings in an oral brief. Trainees may use visual aids such as charts, maps, whiteboard, etc. (150 minutes) **Output:** The trainee must achieve 5/7 SATs to pass this EC. **Feedback:** Instructor will debrief trainees individually following completion of the EC.

Resources	References
1) CSNI lab 2) Secret standalone workstation with access to printer	Nil.

Notes to Instructor
• Multiple trainees can be assigned the same threat platform and research can be completed as a group, however, the EW briefing note must be drafted individually.

Limitations	Nil.
Impact of Failure	IAW NPTORD 4500-54.

Item	EC 001.04 PREPARE EW CORRESPONDENCE	SAT	UNSAT	Remarks
1. Prepare EW briefing note				
Presentation included the following on Surface platform; - Country - Platform length - Number of active platform - Hull distribution per bases - Bases locations on map - Platform weapons - Platform sensors - Weapons range - Weapons speed - ASM ZIPPO - Standoff distance - EPM recommendation - HPD recommendation		☐	☐	
Presentation included the following on Air platform; - Country - Platform maximum speed - Platform maximum range - Platform combat range - Number of active platform - Platform distribution per bases - Bases location on map - Platform weapons - Platform sensors - Weapons range - Weapons speed - EPM recommendation - HPD recommendation		☐	☐	
Presentation included an assessment on the following; - Threat posed by platform - Enemy tactics - Enemy sensors employment - Enemy weapons employment		☐	☐	

Presentation included a recommendation on the following; - Own sensor employment - Own weapons employment - Tactics employment	☐	☐	
Information provided is accurate IAW DSI Threat Web	☐	☐	
Met time requirement for briefing (no more than 10 minutes)	☐	☐	
Delivery: - Military appearance - Professional bearing - Clear voice/Tone - Avoided distracting mannerisms	☐	☐	

Score: /7
Trainee must achieve 5/7 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the trainee.		**Trainee** I have been debriefed on the result of this assessment.	
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

PROTECTED B (when complete)

NTDC(A) Approval – 26/01/2021
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Version No: 3.0
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EO 001.05 – Coordinate EW Operational Training

001.05.01 – Training Initiatives

Conditions		Sequence
Classroom & SEACOT		After EO 001.04
Time	**Classification**	**Instructor**
2100 minutes	SECRET	NES Op PO1/PO2

Learning Objectives	Method
1) Plan/design training events 2) Produce OCS/TABORD message for serial planning 3) Input training scenarios 4) Update training scenarios; 5) Assist in individual training events 6) Assess individual training performance 7) Assist in team training events 8) Assess team training performance 9) Conduct individual training events; 10) Conduct team training events 11) Conduct Unit On Job Training (OJT), Ships Development Package (SDP), Pre-Information Package (PIP)	• Self-Study • Guided Discussion • Practical • Interactive Lecture

Activity Steps	
	<u>LO 1&2: Self Study & Guided Discussion (300 minutes)</u> 1) Trainees will conduct preliminary readings relating to the training process from the TDC(A) PO 001 Required Reading List 2) Upon completion of reading, the instructor will conduct a guided discussion on the following subjects: a. Combat Requirement Readiness (CRR) b. EW education and training; c. The Sea Training Guide Content d. EWX's 3) Examples of questions include: a. What is the purpose of Combat Readiness Requirements? b. What publication would information be found related to CRR's? c. What Electronic Warfare Exercises have you experiences and what was involved? d. What process do we use in the RCN to verify trade specific requirements and ability onboard HMCS ships? (Think periodic assessment) <u>LO 3 & 4 Practical (200 minutes)</u> 1) The instructor will select an EW exercise from AXP-2 for each trainee. Trainees will draft an OCS/TABORD message to meet training objectives outlined in AXP-2.

	2) Trainees will be tasked to draft realistic EW Exercise scripts to include; a. Short scenario b. Sequence of events c. Events description d. Timings 3) Scripts drafted by students shall include a minimum of 3 consorts and have their unit acting as OCS for the serial. 4) Once completed, students will outline the details of their scripts to the class on the whiteboard. The instructor providing feedback on the validity of the scenarios and make recommendations as necessary. <u>LO 5 to 11: Practical (1600 minutes)</u> 5) Trainees will execute the serials outlined in the TABORDs they created during LO 3 & 4. This will be conducted in a simulated operational environment and emphasis will be on planning, voice reporting and net discipline. Some serials will require participation from multiple units, therefore, it is expected that trainees will be designated to perform support roles for each exercise being performed. 6) Upon completion of all serials, trainees are expected to review and make adjustments, as necessary, to their TABORDs to enhance the validity of their training scenario.

3

Resources	References
1) Standalone computer with access to printer 2) Message drafting software 3) Access to printer 4) SEACOT or SEAWOLF 5) TRINITY website 6) NTDC(A) PO 001 Required Reading List	1) AXP-2 2) APP-11 3) APP-1 4) CFCD-106 5) CFCD-102 6) CFCD-114 7) Sea Training Guide

Notes to Instructor

- Trainees should act as operators and role play in classmate's serials for added realism.

EC 001.05 – Coordinate EW Operational Training

Scope	Conditions	Sequence
EO 001.05	Classroom/SEACOT	After EO 001.01
Time	**Classification**	**Instructor**
700 minutes	SECRET	NES Op PO1/PO2

Assessment Steps	
	Input: Trainees will be provided with a copy of EC 001.05 checklist, be given time to review and ask clarifying questions as required. Trainees will use the serial they each produced during EO 001.05 LO 1&2/ **Performance:** Trainees will conduct their serials in a simulated operational environment with focus on planning, voice reporting and net discipline. They are required to conduct a short debrief to include areas for improvement upon serial completion **Output:** Trainee must be successful in all sections to pass this EC. The trainee must achieve: 1) Section I: 12/15 SATs 2) Section II: 5/6 SAT **Feedback:** Instructor will debrief trainees individually following completion of the EC.

Resources	References
1) SEACOT/SEAWOLF 2) LO 1/2 OCS/TABORD messages	1) AXP-2 2) APP-11 3) APP-1 4) CFCD-106 5) CFCD-102 6) CFCD-114 7) Sea Training Guide

Notes to Instructor

- Trainees will be assessed individually on the conduct and execution of their serial.
- Non-assessed trainees may be used as required for scenario realism to improve training value.
- Each trainee will be assessed as OCS for own serial

Limitations	Nil.
Impact of Failure	IAW NPTORD 4500-54.

Item	EC 001.05 Coordinate EW Operational Training	SAT	UNSAT	Remarks
1. OCS/TABORD IAW AXP-02 Standard Order Table				
Line A – Identifying number of exercise and short title		☐	☐	
Line B – Officer scheduling the exercise identified		☐	☐	
Line C – Officer in tactical command identified		☐	☐	
Line D – Officer conducting the serial identified		☐	☐	
Line F – Participating units identified		☐	☐	
Line H – Zone time to be used identified		☐	☐	
Line J – Date and time the exercise is to start identified		☐	☐	
Line JJ – Date and time the exercise is to stop identified		☐	☐	
Line N – Additional information such as; - Rendez-vous - Exercise area - Particular navigational hazards - Other details		☐	☐	
Line ST – Special instructions details the following; - Conduct - Pre-serial brief time and communication line - Sequence of events		☐	☐	
Line U – Communications instructions identified		☐	☐	
Line W – Duty instructions identified		☐	☐	
Line Y – Standard EW instructions identified IAW AXP-02 table 2A-1		☐	☐	
Line Z – Movements on completion identified		☐	☐	
Line ZZ – Additional information such as; - Record keeping and/or analysis instructions		☐	☐	

- Acknowledge requirement			
2. Serial Execution			
Communication check conducted	☐	☐	
Pre-serial brief conducted and included the following; - Sequence of event - Conduct description - Post serial brief intentions	☐	☐	
Execution of exercise IAW AXP-02	☐	☐	
EW exercise meets training requirements as detailed in AXP-02	☐	☐	
Serial debrief conducted	☐	☐	
Voice communication IAW APP-01 throughout exercise conduct	☐	☐	

Score:
Section I /15
Section II /6
Trainee must be successful in all sections to pass this EC. The trainee must achieve:
1) Section I: 12/15 SATs
2) Section II: 5/6 SAT

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the trainee.		**Trainee** I have been debriefed on the result of this assessment.	
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

PROTECTED B (when complete)

PC 001 – Prepare Electronic Warfare Correspondence

Scope	Conditions	Sequence
EO 001.01 to 001.05	Classroom	After EC 001.05
Time	**Classification**	**Instructor**
350 minutes	SECRET	NES OP PO1/PO2

Scenario	To be provided by NTDC(A).
Assessment Steps	**Input:** Trainees will be provided with a copy of PC 001 checklist, be given time to review and ask clarifying questions as required. Trainees will be provided with: 1) NTDC(A) approved scenario 2) Threat sheets and Order of Battle (produced during EO 001.04) 3) Standing OPTASK messages 4) Blank EMCON table **Performance:** Trainees will produce 1 OPTASK EW SUPP and then 1 EMCON table. **Output:** Trainee must be successful in all sections to pass this PC. The trainee must achieve: 1) Section I: 17/22 SATs 2) Section II: 8/10 SATs 3) Section III: 1/1 SAT 4) Section IV: 3/4 SATs 5) Section V: 1/1 SAT **Feedback:** Instructor will debrief trainees individually following completion of the PC.

Resources	References
1) Secret standalone workstation with access to printer 2) Message drafting software 3) Threat sheets created during EO 001.04 4) OOB 5) EMCON table 6) Standing OPTASK messages 7) NTDC(A) approved PC 001 scenario	1) ATP-01 Vol 1 2) ATP-31 3) APP-11 4) CFCD-114 5) CFCD-106

Notes to Instructor
Nil.

Limitations	PC 001 will be conducted as an individual assessment.
Impact of Failure	IAW NPTORD 4500-54.

Item	PC 001 PREPARE OPTASK EW SUPP & PREPARE EMCON TABLE	SAT	UNSAT	Remarks
1. OPTASK EW SUPP IAW APP-11				
Message identification - Exercise nickname - Exercise identifier		☐	☐	
Operation codeword - Operation codeword - Plan originator and number - Option nickname - Secondary option nickname		☐	☐	
Message identifier - Message text format identifier - Standard - Version - Originator - Message serial number - Reference time of publication - Qualifier - Serial number of qualifier - Message security policy - Message security classification - Message security category		☐	☐	
References - Serial letter - Communication type - Originator - Date and/or time of reference - Reference serial number - Special notation - SIC or file number		☐	☐	
Period of time - Start day-time - Stop day-time - Qualified stop time		☐	☐	
Conduct of operations - Text indicator - Instructions present (free text)		☐	☐	
Enemy electronic order of battles - Priority identifier - Emitter category		☐	☐	

- Emitter designation, nickname or spot number - Nation of emitter			
Target of interest - Priority identifier - Emitter category - Radar frequency - Pulse repetition frequency - Pulse repetition interval - Pulse width - Scan period - Nation of emitter	☐	☐	
Duty assignment details - Unit or organization - Duty code	☐	☐	
Electronic support measures tasking - Equipment designator - Frequency limits or spot number - Unit	☐	☐	
Tasking of jammers - Platform carrying equipment - Jamming control authority for unit - Jammer type to be used - Frequency limits or spot number - Jammer alert state	☐	☐	
CHAFF employment instructions - Text indicator - Instructions present (free text)	☐	☐	
Emission control plan format - Index number or letter - Equipment or tactical unit	☐	☐	
Emission control plan details - Emission control plan details - Index letter(s) - Equipment index and radiation status indicator	☐	☐	
Force electronic orders of battles - Unit name - Correction to EOB - EW equipment type	☐	☐	
Prevention of mutual interference instructions - Text indicator - Instructions present (free text)	☐	☐	

Satellite countermeasures instructions - Text indicator - Instructions present (free text)	☐	☐	
Satellite vulnerability details - Satellite system name or designator - Start time of satellite vulnerability - Stop time of satellite vulnerability	☐	☐	
Reporting instructions - Text indicator - Instructions present (free text)	☐	☐	
Special instructions - Text indicator - Instructions present (free text)	☐	☐	
Special information - Text indicator - Instructions present (free text)	☐	☐	
Acknowledge requirement - Acknowledge requirement indicator - Instructions for force/unit required to acknowledge	☐	☐	
2. General EMCON Considerations Taken IAW ATP-01 Vol 1			
Reflect/support OTC/CWC aim	☐	☐	
Accounts for enemy threat	☐	☐	
Accounts for enemy capabilities	☐	☐	
Accounts for enemy tactics	☐	☐	
Cover all aspects of warfare including; - Electro-Optical - Bridge dial illumination - Cigarette glows considerations - De-icing equipment - Funnel exhaust - Signal projector - Radiated acoustic noise - Operation of auxiliary machinery - Non-essential crew activity	☐	☐	
Provide sufficient time warning for the use of pickets	☐	☐	
Allow for reaction time of sensors and weapons that have been silent	☐	☐	
Allow for ROE in use	☐	☐	
Reflect the disposition of the following; - Maritime units	☐	☐	

- Areas of operations of supporting A/C - Surface ships - Submarines			
Allow for the following; - Periodic system check - Periodic system maintenance	☐	☐	
3. Tactical Situation Considerations Taken IAW ATP-01 Vol 1			
Table has one specific plan for each of the following TACSIT; - Unknown - Force located and targeted - Force location known; disposition unknown - Force not located	☐	☐	
4. EMCON Objectives Met IAW ATP-01 Vol 1			
Gain maximum information from enemy	☐	☐	
Deny information to enemy	☐	☐	
Includes all emitters	☐	☐	
Easy to understand	☐	☐	
5. Radar Status Indicator Employment IAW ATP-01 Vol 1			
RSI employed IAW ATP-01 Vol 1	☐	☐	

Score:
Section I /22
Section II /10
Section III /1
Section IV /4
Section V /1

Trainee must be successful in all sections to pass this PC. The trainee must achieve:

1) Section I: 17/22 SATs
2) Section II: 8/10 SATs
3) Section III: 1/1 SAT
4) Section IV: 3/4 SATs
5) Section V: 1/1 SAT

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the trainee.		**Trainee** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

PROTECTED B (when complete)

PO 002 – Operate Electronic Warfare Systems

1. Performance: Operate Electronic Warfare Systems.

2. Conditions:

   a. Given:
      (1) Command direction,
      (2) Supervision,
      (3) Electro-Optical Detection System,
      (4) Infra-Red Detection System,
      (5) Electronic Attack System,
      (6) Shipborne Decoy System,
      (7) Communication Terminal,
      (8) Multi-Function Workstation,
      (9) CSNI access, and
      (10) References,

   b. Denied: nil; and

   c. Environmental:
      (1) Day or night, and
      (2) At sea or alongside.

3. Standard: the EWS shall operate Electronic Warfare Systems IAW references by:

   a. Operating Combat System equipment (B00080) by:
      (1) Configuring Multi-Function Workstation (MFW) for operations,
      (2) Manipulating MFW and associated menus/sub-menus, and
      (3) Performing pre-sail checks on equipment (B00170),

   b. Operating AAW/ASuW Weapons and Sensors (D0060 & D00010) to include:
      (1) Operating Electro-Optical (EO) and Infra-Red (IR) Detection Systems (C00200/C00440) by:
         (a) Configuring EO/IR detection systems for operations (C00100/C00150), and
         (b) Manipulating EO/IR detection systems and associated menus/sub-menus,

      (2) Conducting Electronic Attacks (C00020) by:
         (a) Configuring EA system for operations, and
         (b) Manipulating EA systems and associated menus/sub-menus, and

   c. Configuring the communication terminal for operation (A00030) by:

(1) Manipulating communication terminal system menus, and
(2) Joining communication nets.

4. Training Limitations: Nil.

EO 002.01 – Operate Combat System Equipment

002.01– Introduction to MFW

Conditions		Sequence
Classroom & SEACOT/SEAWOLF		Start of course
Time	**Classification**	**Instructor**
1200 minutes	SECRET	NES Op PO1/PO2

Learning Objectives	Method
1) Define MFW and associated menus/sub-menus 2) Manipulate MFW and associated menus/sub-menus 3) Configure MFW for operations 4) Perform pre-sail checks on equipment	• Self-Study • Guided Discussion • EDI

Activity Steps	
	<u>LO 1: Self-Study & Guided Discussion (400 minutes)</u> 1) Trainees will conduct preliminary readings relating to CMS330 and the MFW from the NTDC(A) PO 002 Required Reading List. 2) Following completion of reading, the instructor will conduct a guided discussion on the following subjects: a. System overview b. User interface functional description c. AWW sensors <u>LO 2 to 5: EDI (800 minutes)</u> 3) The instructor will project their MFW display and use the NTDC(A) provided MFW Overview checklist to conduct a MFW general summary and provide a brief explanation on each MFW display, menu and sub-menu. 4) Trainees will follow each step on their own MFW. 5) The instructor will demonstrate the basics (defined in the NTDC(A) Operator Function checklist) of how to manipulate the MFW in the EWS role. 6) Trainees will then practice system manipulation. 7) The instructor will demonstrate how to setup the MFW in the EWS role using the NTDC(A) provided EWS Set-Up checklist. 8) Trainees will then practice the setup.

Resources	References
1) Secret standalone workstation 2) NTDC(A) EWS Set-Up checklist 3) NTDC(A) MFW Overview checklist 4) NTDC(A) System Manipulation checklist 5) SEACOT/SEAWOLF MFW 6) Projector	1) CSUMs

Notes to Instructor

Nil.

EC 002.01 – Operate Combat System Equipment

Scope	Conditions	Sequence
EO 002.01	SEACOT/SEAWOLF	After EO 002.01
Time	**Classification**	**Instructor**
150 minutes	SECRET	NES OP PO1/PO2

Assessment Steps	
	Input: Trainees will be provided with a copy of EC 002.01 checklist, be given time to review and ask clarifying questions as required. The instructor will: 1) Use the EWS Set Up checklist to ensure trainees meet the criteria while setting up EWS 2) Ask a minimum of 5 questions from the guided discussions covered during EO 002.01 to meet the assessment factors of the EC checklist 3) Select 5 system manipulation actions for each trainee to perform **Performance:** Trainees will be assessed individually on: 1) Conducting EWS set-up based on the NTDC(A) EWS Set-Up checklist used during EO 002.01 2) Answering questions covered during EO 002.01 guided discussions 3) Executing operator functions in the EWS role **Output:** Trainee must be successful in all sections to pass this EC. The trainee must achieve: 1) Section I: 4/5 SATs 2) Section II: 3/4 SATs **Feedback:** Instructor will debrief trainees individually following completion of the EC.

Resources	References
1) SEACOT/SEAWOLF MFW	1) CSUMs

Notes to Instructor

- Trainees are not allowed any of the previously provided equipment checklists during this EC.
- System manipulations and questions chosen are to be annotated by the instructor in the EC checklist.

Limitations	Nil.
Impact of Failure	IAW NPTORD 4500-54.

Item	EC 002.01 OPERATE COMBAT SYSTEM EQUIPMENT	SAT	UNSAT	Remarks
1. TACPLOT				
Speed leaders		☐	☐	
Radar video		☐	☐	
Charts		☐	☐	
Range rings		☐	☐	
Mode 4		☐	☐	
2. Configure/Manipulate MFW for Operations				
Fusion Settings - EW manual - CIS - ESM - SIRIUS - Auto upgrade/drop		☐	☐	
Tactical Decision Aids (TDA) - POP UP - FU/FT/FI		☐	☐	
SID		☐	☐	
Filters		☐	☐	

Score:
Section I /5
Section II /4
Trainee must be successful in all sections to pass this EC. The trainee must achieve:
1) Section I: 4/5 SATs
2) Section II: 3/4 SATs

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**

Assessor I have discussed the results of this assessment with the trainee.		**Trainee** I have been debriefed on the result of this assessment.	
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

PROTECTED B (when complete)

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EO 002.02 – Operate AAW/ASuW Weapons and Sensors

002.02.01 – Introduction to EO/IR Sensors

Conditions		Sequence
Classroom & SEACOT/SEAWOLF/Ship		After EO 002.01
Time	**Classification**	**Instructor**
900 minutes	SECRET	NES Op PO1/PO2

Learning Objectives	Method
1) Operate EO and IR Detection Systems 2) Define EO/IR detection systems and associated menus/sub-menus 3) Manipulate EO/IR detection systems and associated menus/sub-menus 4) Configure EO/IR detection systems for operations	• Self-Study • Guided Discussion • EDI

Activity Steps	
	<u>LO 1: Self-Study & Guided Discussion (300 minutes)</u> 1) Trainees will read the information on the EO/IR systems and its functions from the NTDC(A) PO 002 Required Reading List. 2) Following completion of reading, the instructor will conduct a guided discussion on the following subjects: a. System overview b. System capabilities and limitations c. System functional description d. RCN system employment doctrine <u>LO 2 & 3: EDI (600 minutes)</u> 3) The instructor will project their MFW display and use the NTDC(A) provided EO/IR System Overview checklist to conduct a MFW general summary and provide a brief explanation on each MFW display, menu and sub-menu. 4) Trainees will follow each step on their own MFW. 5) The instructor will demonstrate the basics (defined in the NTDC(A) EO/IR System Operator Function checklist) of how to manipulate the MFW in the EWS role. 6) Trainees will then practice system manipulation.

Resources	References
1) Secret standalone workstation 2) NTDC(A) CMS Set-Up checklist 3) NTDC(A) EO/IR System Overview checklist 4) NTDC(A) EO/IR System Operator Function checklist	1) ATP-31 2) CSUMs 3) U MAN-1120 SIRIUS manual 4) CFCD-106 5) CFCD-114

5) SEACOT/SEAWOLF trainer or ship 6) Projector 7) NTDC(A) PO 002 Required Reading List	

Notes to Instructor

Nil.

EO 002.02 – Operate AAW/ASuW Weapons and Sensors

002.02.02 – Introduction to EA Systems

Conditions		Sequence
Classroom & SEACOT/SEAWOLF/Ship		After EO 002.02.01
Time	**Classification**	**Instructor**
600 minutes	SECRET	NES Op PO1/PO2

Learning Objectives	Method
1) Conduct Electronic Attacks (EA) 2) Define EA system and associated menus/sub-menus 3) Manipulate EA system and associated menus/sub-menus 4) Configure EA system for operations	• Self-Study • Guided Discussion • EDI

Activity Steps	
	<u>LO 1: Self-Study & Guided Discussion (300 minutes)</u> 1) Trainees will read the information on the EA system and its functions from the NTDC(A) PO 002 Required Reading List. 2) Following completion of reading, the instructor will conduct a guided discussion on the following subjects: a. System overview b. System capabilities and limitations c. System functional description d. RCN system employment doctrine <u>LO 2 & 3: EDI (300 minutes)</u> 3) The instructor will project their MFW display and use the NTDC(A) provided EA System Overview checklist to give an EA system general summary with all trainees and provide a brief explanation of each display, menus, sub-menus and buttons within the MFW and on the system. 4) Trainees will follow each step on their own MFW. 5) The instructor will demonstrate the basics (defined in the NTDC(A) EO/IR System Operator Function checklist) of how to manipulate the MFW in the EWS role. 6) Trainees will then practice system manipulation.

Resources	References
1) Secret standalone workstation 2) NTDC(A) CMS Set-Up check list 3) NTDC(A) EA System Overview check list 4) NTDC(A) EA System Operator Function check list 5) SEACOT/SEAWOLF trainer or ship 6) Projector	1) CSUMs 2) EA system user manual 3) CFCD-106 4) CFCD-114

7) NTDC(A) PO 002 Required Reading List	

Notes to Instructor

Nil.

EO 002.02 – Operate AAW/ASuW Weapons and Sensors

002.02.03 – Introduction to Shipborne Decoy Systems

Conditions		Sequence
Classroom/SEACOT/SEAWOLF & Ship/NAD		After EO 002.02.01
Time	**Classification**	**Instructor**
1000 minutes	SECRET	NES Op PO1/PO2

Learning Objectives	Method
1) Operate Shipborne Decoy System 2) Define shipborne decoy systems and associated menus/sub-menus 3) Manipulate shipborne decoy systems and associated menus/sub-menus 4) Configure shipborne decoy systems for operations 5) Perform the shipborne decoy system supervisor duties	• Self-Study • Guided Discussion • EDI • Practical

Activity Steps	
	<u>LO 1: Self-Study & Guided Discussion (300 minutes)</u> 1) Trainees will read the information on the shipborne decoy systems and their functions from the NTDC(A) PO 002 Required Reading List. 2) Following completion of reading, the instructor will conduct a guided discussion on the following subjects: a. System overview b. System capabilities and limitations c. System functional description d. RCN system employment doctrine <u>LO 2 & 3: EDI (500 minutes)</u> 3) The instructor will project their MFW display and use the NTDC(A) provided Shipborne Decoy and DLF Systems Overview checklist to provide a decoy systems general summary then provide a brief explanation on each display, menus, sub-menus, buttons and levers within the MFW and on the decoy systems themselves. 4) Trainees will follow each step on their own MFW. 5) The instructor will demonstrate the basics (defined in the NTDC(A) Decoy Systems Operator Function and the DLF Launch Procedure checklist) of how to manipulate the decoy systems in the EWS role. 6) Trainees will then practice system manipulation. <u>LO 4 & 5: Practical (200 minutes)</u> 7) Trainees are to give a safety brief to include safe ammunition handling and loading procedures. 8) Then trainees are to demonstrate their ability to supervise the load/unload and preparation of the Shipborne Decoy systems, including MASS and DLF.

	9) Students must demonstrate: a. Ability to define and make mechanically/electrically safe for both MASS and DLF b. Knowledge of misfire procedure (including time before unload and how to conduct BIT) c. How to manipulate the MASS Manual Panel to prepare for load/unload procedure d. Knowledge of where the keys are kept IAW policy and how this relates to actions stations, entering/leaving harbour, etc.

Resources	References
1) Secret standalone workstation 2) NTDC(A) CMS Set-Up check list 3) NTDC(A) Decoy Systems Overview checklist 4) NTDC(A) DLF System Overview checklist 5) NTDC(A) Decoy Systems Operator Function checklist 6) SEACOT/SEAWOLF trainer or ship 7) Projector 8) NTDC(A) Required Reading List	1) CSUMs; 2) TRASE2430EW Operational and Technical Manual Draft 3) CFCD-106 4) NCPM 1000

Notes to Instructor

- LO 4 & 5 are to be conducted either at NAD or aboard available ship. System keys are to be available and system is to be manipulated (Safe/Unsafe)
- During LO 4 & 5, trainees are to play the role of the Loading supervisor one at the time. Remaining trainees to alternatively play the loaders role

EO 002.03 – Configure Communication Terminal for Operation Operations
002.03.01 – Introduction to SHINCOM

Conditions		Sequence
Classroom & SEACOT/SEAWOLF		After EO 002.02.03
Time	**Classification**	**Instructor**
350 minutes	SECRET	NES Op PO1/PO2

Learning Objectives	Method
1) Define the communication terminal system and associated menus/sub-menus 2) Manipulate the communication terminal system and associated menus/sub-menus 3) Configure communication terminal system for operations	• Self-Study • Guided Discussion • EDI

Activity Steps	
	<u>LO 1: Self-Study & Guided Discussion (200 minutes)</u> 1) Trainees will read the information on the communication system and its functions from the NTDC(A) PO 002 Required Reading List. 2) Following completion of reading, the instructor will conduct a guided discussion on the following subjects: a. Net management b. Net discipline <u>LO 2 & 3: EDI (150 minutes)</u> 3) The instructor will project their MFW display and use the NTDC(A) provided Communication System Overview checklist to conduct a communication system general summary and provide a brief explanation on each menu, sub-menu and buttons. 4) Trainees will follow each step on their own MFW. 5) The instructor will demonstrate the basics (defined in the NTDC(A) Communication System Operator Function checklist) of how to manipulate the system in the EWS role. 6) Trainees will then practice system manipulation.

Resources	References
1) Secret standalone workstation 2) NTDC(A) CMS Set-Up checklist 3) NTDC(A) Communication System Overview checklist 4) NTDC(A) Communication System Operator Function checklist 5) SEACOT or SEAWOLF trainer 6) Projector 7) NTDC(A) Required Reading List	1) SHINCOM handbook

Notes to Instructor

Nil.

EC 002.03– Operate ASuW Systems and Communications

Scope	Conditions	Sequence
EO 002.02 & 002.03	SEACOT/SEAWOLF	After EO 002.03
Time	**Classification**	**Instructor**
400 minutes	SECRET	NES Op PO1/PO2

Assessment Steps	
	Input: Trainees will be provided with a copy of EC 002.03 checklist, be given time to review and ask clarifying questions as required. The instructor will: 1) Ensure proper setup on: a. EO/IR b. EA c. Decoy System d. Communication Systems 2) Ask a minimum of 20 questions from the guided discussions covered during EO 002.02 & 002.03 to meet the assessment factors of the EC checklist 3) Select 20 system manipulation actions for each trainee to perform **Performance:** Trainees will be assessed individually on: 1) Conducting EO/IR, EA, Decoy System and Communication Systems setup 2) Answering questions covered during EO 002.02 & 002.03 guided discussions 3) Executing operator functions in the EWS role **Output:** Trainee must be successful in all sections to pass this PC. The trainee must achieve: 1) Section I: 8/10 SATs 2) Section II: 4/5 SATs 3) Section III: 5/6 SAT 4) Section IV: 2/2 SATs 5) *Section V: 4/4 SATs 6) **Section VI: 4/4 SATs *For each system within Section V, to obtain a SAT trainees must successfully answer 4/5 questions correctly. ** For each system within Section VI, to obtain a SAT trainees must successfully perform 4/5 actions correctly **Feedback:** Instructor will debrief trainees individually following completion of the EC.

Resources	References

1) SEACOT/SEAWOLF MFW	1) CSUMs 2) EO/IR system user manual 3) EA system user manual 4) Decoy system user manual 5) SHINCOM handbook

Notes to Instructor

- Trainees are not allowed any of the previously provided equipment checklists during this EC.
- System manipulations and questions chosen are to be annotated by the instructor in the EC checklist.

Limitations	Nil.
Impact of Failure	IAW NPTORD 4500-54.

Item	EC 002.03 OPERATE ASuW SYSTEMS AND COMMUNICTIONS	SAT	UNSAT	Remarks
1. Conduct EO/IR System Set-Up				
Operating modes selected		☐	☐	
Dynamic variable		☐	☐	
FBA targets		☐	☐	
Ancillary zones		☐	☐	
Video feeds 1 thru 4 set-up		☐	☐	
MW/LW IR		☐	☐	
Blanking sector		☐	☐	
Met data		☐	☐	
Fault isolation		☐	☐	
Day/night		☐	☐	
2. Conduct EA System Set-Up				
HV On selected		☐	☐	
Auto selected		☐	☐	
Fault isolation		☐	☐	
Jam pre sets		☐	☐	
Jam immediate pre sets		☐	☐	
3. Conduct Decoy Systems Set-Up				
MFD powered on		☐	☐	
Verify interface status		☐	☐	
BIT and launchers check		☐	☐	
Load threat		☐	☐	
Input own ship crs and spd		☐	☐	
MASS zones displayed		☐	☐	
4. Conduct Communication System Set-Up				
Internet Net		☐	☐	
External Net		☐	☐	
5. Answer Confirmation Questions				
EO/IR system (answer 4/5 correctly to get SAT) (1) (2) (3)		☐	☐	

(4) (5)			
EA system (answer 4/5 correctly to get SAT) (1) (2) (3) (4) (5)	☐	☐	
Decoy systems (answer 4/5 correctly to get SAT) (1) (2) (3) (4) (5)	☐	☐	
Communication system (answer 4/5 correctly to get SAT) (1) (2) (3) (4) (5)	☐	☐	
6. Execute Operator Function Manipulations			
EO/IR system (answer 4/5 correctly to get SAT) (1) (2) (3) (4) (5)	☐	☐	
EA system (answer 4/5 correctly to get SAT) (1) (2) (3) (4) (5)	☐	☐	
Decoy systems (answer 4/5 correctly to get SAT) (1) (2) (3) (4)	☐	☐	

(5)			
Communication system (answer 4/5 correctly to get SAT) (1) (2) (3) (4) (5)	☐	☐	

Score:
Section I /10
Section II /5
Section III /6
Section IV /2
Section V /4
Section VI /4

Trainee must be successful in all sections to pass this EC. The trainee must achieve:

1) Section I: 8/10 SATs
2) Section II: 4/5 SATs
3) Section III: 5/6 SATs
4) Section IV: 2/2 SATs
5) Section V: 4/4 SATs
6) Section VI: 4/4 SATs

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the trainee.		**Trainee** I have been debriefed on the result of this assessment.	
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

PROTECTED B (when complete)

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PC 002 – Operate Electronic Warfare Systems

Scope	Conditions	Sequence
EO 002.01 to 002.03	SEACOT/SEAWOLF	After EC 002.03
Time	**Classification**	**Instructor**
400 minutes	SECRET	NES Op PO1/PO2

Assessment Steps	
	Input: Trainees will be provided with a copy of PC 002 checklist, be given time to review and ask clarifying questions as required. The instructor will: 1) Ask 5 questions on each system from the guided discussions covered during EO 002.01 to 002.03 2) Select 5 operator function actions on each system for each trainee to perform **Performance:** Trainees will respond to questions asked by the instructor and perform operator function actions for each of the following systems: 1) MFW 2) EO/IR 3) EA 4) Decoy 5) Communication **Output:** Trainee must be successful in all sections to pass this PC. The trainee must achieve: 1) Section I: 4/5 SATs 2) Section II: 8/10 SATs 3) Section III: 4/5 SATs 4) Section IV: 4/6 SAT 5) Section V: 2/2 SATs 6) *Section VI: 5/5 SATs 7) **Section VII: 5/5 SATs *For each system within Section V, to obtain a SAT trainees must successfully answer 4/5 questions correctly. ** For each system within Section VI, to obtain a SAT trainees must successfully perform 4/5 actions correctly **Feedback:** Instructor will debrief trainees individually following completion of the PC.

Resources	References
1) SEACOT/SEAWOLF trainer	1) CSUM 2) EO/IR System user manual 3) EA System user manual 4) Decoy System user manuals 5) SHINCOM Handbook

Notes to Instructor

- Trainees shall not be in possession of the CMS Set-Up check list during this PC
- System operator functions as well as system specific questions have to be annotated by the instructor in the PC checklist

Limitations	PC 002 will be conducted as an individual assessment.
Impact of Failure	IAW NPTORD 4500-54.

Item	PC 002 OPERATE ELECTRONIC WARFARE SYSTEMS	SAT	UNSAT	Remarks
1. Conduct Full EWS Set-Up				
TACPLOT - Speed leaders - Radar video - Charts - Range rings - Mode 4		☐	☐	
Fusion Settings - EW manual - CIS - ESM - SIRIUS - Auto-upgrade/drop		☐	☐	
Tactical Decision Aids - Pop-up - FU/FT/FI		☐	☐	
SID		☐	☐	
Filters selected		☐	☐	
2. EO/IR System Set-Up				
Operating mode selected		☐	☐	
Dynamic variable selected		☐	☐	
FBA Targets		☐	☐	
Ancillary zones		☐	☐	
Video feeds 1 thru 4		☐	☐	
MW/LW IR		☐	☐	
Blanking sectors		☐	☐	
MET data		☐	☐	
Fault isolation		☐	☐	
Day/night		☐	☐	
3. EA System Set-Up				
HV on selected		☐	☐	
Auto selected		☐	☐	
Fault isolation		☐	☐	

Jam pre-sets	☐	☐	
Jam immediate pre-sets	☐	☐	
4. Decoy Systems Set-Up			
MFD powered on	☐	☐	
Interface status verified	☐	☐	
BIT and launcher checks conducted	☐	☐	
Threat files loaded	☐	☐	
Own ship course and speed inputted	☐	☐	
Mass zones displayed	☐	☐	
5. Communication System Set-Up			
Internal net set	☐	☐	
External net set	☐	☐	
6. Answer Confirmation Questions			
MFW system (answer 4/5 correctly to get SAT) (1) (2) (3) (4) (5)	☐	☐	
EO/IR system (answer 4/5 correctly to get SAT) (1) (2) (3) (4) (5)	☐	☐	
EA system (answer 4/5 correctly to get SAT) (1) (2) (3) (4) (5)	☐	☐	
Decoy systems (answer 4/5 correctly to get SAT) (1) (2) (3) (4) (5)	☐	☐	

Communication system (answer 4/5 correctly to get SAT) (1) (2) (3) (4) (5)	☐	☐	
7. Execute Operator Function Manipulations			
MFW system (answer 4/5 correctly to get SAT) (1) (2) (3) (4) (5)	☐	☐	
EO/IR system (answer 4/5 correctly to get SAT) (1) (2) (3) (4) (5)	☐	☐	
EA system (answer 4/5 correctly to get SAT) (1) (2) (3) (4) (5)	☐	☐	
Decoy systems (answer 4/5 correctly to get SAT) (1) (2) (3) (4) (5)	☐	☐	
Communication system (answer 4/5 correctly to get SAT) (1) (2) (3) (4) (5)	☐	☐	

Score:
Section I /5
Section II /10
Section III /5
Section IV /6
Section V /2
Section VI /5
Section VII /5

Trainee must be successful in all sections to pass this EC. The trainee must achieve:

1) Section I: 4/5 SATs
2) Section II: 8/10 SATs
3) Section III: 4/5 SATs
4) Section IV: 4/6 SATs
5) Section V: 2/2 SATs
6) Section VI: 5/5 SATs
7) Section VII: 5/5 SATs

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the trainee.		**Trainee** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

PROTECTED B (when complete)

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PO 003 – Execute AWW Tactics for EW Operations

1. Performance: Execute AWW Tactics for EW Operations.

2. Conditions:

   a. Given:
      (1) Command direction,
      (2) Supervision,
      (3) Electro-Optical Detection System,
      (4) Infra-Red Detection System,
      (5) Electronic Attack System,
      (6) Shipborne Decoy System,
      (7) Communication Terminal,
      (8) Multi-Function Workstation, and
      (9) References,

   b. Denied: nil; and

   c. Environmental:
      (1) Day or night, and
      (2) At sea or alongside.

3. Standard: the EWS shall execute AWW Tactics for EW Operations IAW references by:

   a. Coordinating Local EW operations (C00300) by:
      (1) Interpreting Electro-Optic (EO) process,
      (2) Interpreting Infra-Red (IR) process,
      (3) Interpreting Electronic Intelligence (ELINT) concepts,
      (4) Interpreting/Applying Electronic Attack (EA) methods, and
      (5) Interpreting/Applying Electronic Protect (EP) procedures,

   b. Operating EW Equipment (C00030) by:
      (1) Optimizing EW sensors employment (C00660),
      (2) Detecting radar contacts (B00100),
      (3) Supervising EW sub-team,
      (4) Classifying radar contacts (B00010),
      (5) Tracking electronic intercepts (C00640),
      (6) Associating electronic intercept information (C00060),
      (7) Performing hull to emitter correlations (HULTEC) (C00370),
      (8) Disseminating electronic support (ES) information (C00400),
      (9) Correlating intercept information (F00170), and
      (10) Executing Overt/Covert EW collection operations (C00620),

c. Directing ships ES, EA and EP Operations (C00520) by:
   (1) Coordinating ships ES, EA, EP operations (C00490),
   (2) Performing Electronic Protection (EP) measures (C00070),
   (3) Assessing jamming (C00050),
   (4) Performing electronic deception (C00720), and
   (5) Performing intelligence gathering (B00070),

d. Communicating using tactical voice procedures (A00060) by:
   (1) Performing communication checks (A00020),
   (2) Utilizing EW block reports,
   (3) Authenticating communication transmissions (A00010),
   (4) Utilizing low grade codes, and
   (5) Applying tactical code words,

e. Assessing Anti-Air Warfare (AAW) Threat Information (E00020) by:
   (1) Applying AAW general tactics to include:
      (a) Applying ZIPPO criteria,
      (b) Categorizing missile groups, and
      (c) Conducting AAW duties,

   (2) Coordinating ASMD reactions by:
      (a) Planning ASMD reactions,
      (b) Executing ASMD reactions, and
      (c) Applying ASMD maneuvering,

   (3) Conducting Detect to Engage (DTE) sequence,
   (4) Employing SOFTKILL,
   (5) Maintaining Recognized Air Picture (RAP), and
   (6) Applying Identification Criteria (ID CRIT) and checkprints,

f. Interpreting ASuW General Tactical Procedures by:
   (1) Interpreting AsuW duties,
   (2) Applying promulgated tactical information as it applies to internal and external organizations (C00040),
   (3) Maintaining Recognized Maritime Picture (RMP) (F00010),
   (4) Utilizing Recognition Confidence Level (RCL),
   (5) Performing Area of Interest (AOI) monitoring (C00450),
   (6) Amplifying track information (F00010), and
   (7) Tracking high interest vessels (C00660), and

g. Applying naval gun fire support procedures by:
   (1) Interpreting spotting procedures,
   (2) Interpreting range safety orders, and
   (3) Manipulating naval gun fire software.

4. Training Limitations: Nil.

EO 003.01 – Coordinate Local EW Operations

003.01.01 – Interpret Electronic Warfare Theory

Conditions		Sequence
Classified Classroom		After PO 001
Time	**Classification**	**Instructor**
350 minutes	SECRET	NES Op PO1/PO2

Learning Objectives	Method
1) Interpret Electro-Optic (EO) process 2) Interpret Infra-Red (IR) process 3) Interpret Electronic Intelligence (ELINT) concepts 4) Interpret/Apply Electronic Attack (EA) methods 5) Interpret/Apply Electronic Protect (EP) procedures	• Self-Study **then** • Trainee Presentation **or** • Guided Discussion

Activity Steps

<u>LO 1 to 5: Self-Study (250 minutes)</u>

If conducting Trainee Presentation:

1) Trainees will each read all of the PO 003 Required Reading List EO 003.01.

If conducting Guided Discussion:

1) Trainees will each be assigned a set of readings from the PO 003 Required Reading List EO 003.01 and prepare a presentation.

then

<u>LO 1 to 5: Trainee Presentation (100 minutes)</u>

2) Trainees will present a summary of readings to the class.

or

<u>LO 1 to 5: Guided Discussion (100 minutes)</u>

2) Following completion of reading, the instructor will conduct a guided discussion on the following subjects:
    a. LO 1 & 2:
        i. Characteristics of EO/IR sensors
        ii. Advantages and disadvantages of EO/IR sensors
    b. LO 3:
        i. Define ELINT
        ii. The role of ELINT in an operational environment
        iii. Other aspects of ELINT (COMINT and SIGINT)
    c. LO 4, 5, & 6:
        i. Define EA/EP and give examples of each
        ii. Advantages and disadvantages of EA/EP
        iii. EA/EP operations onboard HMC Ships

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Resources	References
1) PO 003 Required Reading List 2) SECRET standalone workstation	1) ATP-01 Vol 1 2) ATP-31 3) CFSAS Basic Electronic Warfare Manual 4) CFCD-106 5) CFCD-114 6) Communication ESM handbook

Notes to Instructor

- Selection of Trainee Presentation or Guided Discussion is dependent on number of trainees, if the course is running below optimum Guided Discussion is recommended.

EO 003.02 – Operate EW Equipment

003.02.01 – Applying EW Procedures

Conditions		Sequence
Classified Classroom		After EO 003.01
Time	**Classification**	**Instructor**
400 minutes	SECRET	NES Op PO1/PO2

Learning Objectives	Method
1) Optimize EW sensors employment; 2) Detect radar contacts; 3) Supervise EW sub-team; 4) Classify radar contacts; 5) Track electronic intercepts ; 6) Associate electronic intercept information; 7) Perform hull to emitter correlations (HULTEC); 8) Disseminate electronic support (ES) information; 9) Correlate intercept information; and 10) Execute Overt/Covert ELINT collection operations	• Self-Study • Guided Discussion

Activity Steps

<u>LO 1 to 10: Self-Study & Guided Discussion (400 minutes)</u>

1) Trainees will conduct preliminary readings from the NTDC(A) PO 003 Required Reading List on:
    a. Optimization of EW sensors and their employments
    b. Importance of having the EW equipment tasked and manipulated in the most effective manner
    c. Importance of the EW sub-team
2) Following completion of reading, the instructor will conduct a guided discussion on the following subjects:
    a. LO 1:
        i. Allied Joint EW Doctrine
        ii. Shipboard EW sensor employments
        iii. The threat and how it effects EW sensor employment
    b. LO 2:
        i. Detection range of shipboard sensors
        ii. Operators involved in the Radar contact detection process
        iii. Reporting process upon detection of radar contacts
    c. LO 3 to 10:
        i. Supervision of EW sub-teams
        ii. The management of Radar and ESM contacts as they are:
            1. Classified
            2. Tracked
            3. Associated
        iii. Platform Identification based on emitter
        iv. EW information flow and correlation

	v. Executing overt/covert ELINT collection

Resources	References
1) Secret standalone workstation 2) NTDC(A) PO 003 Required Reading List	1) ATP-01 Vol 1 2) ATP-31 3) CFCD-106 4) CFCD-114

Notes to Instructor

Nil.

EO 003.03 – Directing Ships EA/EP/ES Operations

003.03.01 – Directing Ships EA/EP/ES Operations

Conditions		Sequence
Classified Classroom		After EO 003.02
Time	**Classification**	**Instructor**
350 minutes	SECRET	NES Op PO1/PO2

Learning Objectives	Method
1) Coordinate Ships EA/EP/ES Operations 2) Perform electronic protection measures 3) Assess jamming 4) Perform electronic deception 5) Perform intelligence gathering	• Self-Study • Guided Discussion • PowerPoint

Activity Steps	
	<u>LO 1 to 5: Self-Study & Guided Discussion (350 minutes)</u> 1) Trainees will conduct preliminary readings from the NTDC(A) PO 003 Required Reading List on: a. Coordination of EA/EP/ES b. Importance of having Electronic Deception equipment and intelligence gathering capability 2) Upon completion of reading, the instructor will present an Electronic Deception and Intelligence PowerPoint, followed by a guided discussion on the following subjects: a. LO 1 & 2: i. Definition of Electronic Protection and how it is carried onboard HMC Ships ii. Jamming procedures b. LO 3 to 5: i. Application of Electronic Deception ii. Examples of Electronic deception equipment iii. Intelligence gathering process

Resources	References
1) Secret standalone workstation 2) NTDC(A) PO 003 Required Reading List 3) Electronic Deception and Intelligence Gathering PowerPoint	1) ATP-01 Vol 1 2) ATP-31 3) CFCD-106 4) CFCD-114

Notes to Instructor
Nil.

EO 003.04 – Communicate Using Tactical Voice Procedures

003.04.01 – Communicate Using Tactical Voice Procedures

Conditions		Sequence
Classified Classroom		After EO 003.03
Time	**Classification**	**Instructor**
500 minutes	SECRET	NES Op PO1/PO2

Learning Objectives	Method
1) Utilize EW tactical voice procedures 2) Perform communication checks 3) Authenticate communication transmissions 4) Utilize low grade codes 5) Apply tactical code words	• Self-Study • Guided Discussion • Think-Pair-Share • Fishbowl

Activity Steps	
	<u>LO 1 to 5: Self-Study & Guided Discussion (100 minutes)</u> 1) Trainees will conduct readings relating to tactical communication procedures and voice reports employed by the EWS. 2) Following completion of reading, the instructor will conduct a guided discussion on the following subjects: a. Types of communications b. Communication Security (COMSEC) c. Types of EW reports d. AWW codewords/prowords e. ASMD procedures on tactical AAW nets <u>LO 3 & 4 (10 minutes)</u> 3) Trainees will write down everything they know about authenticating communication transmissions and utilizing low grade codes. They will then get with a partner and consolidate their knowledge. 4) One person from each group will present their findings to the class. 5) Instructor will provide amplifying information as required. <u>LO 1 to 5 (390 minutes)</u> 6) Instructors will divide the class into syndicates (3 to 4 trainees per syndicate). Each syndicate will receive a copy of NTDC(A) Tactical Voice Procedures/Codewords List. 7) Syndicates will then take turns leading voice reporting with the remainder of the classroom acting as consorts. 8) Mistakes in voice reporting are to be identified and corrected by the lead syndicate.

Resources	References
1) NTDC(A) Tactical Voice Procedures/Codewords List 2) TDC (A) PO 003 Required Reading List 3) Secret standalone Computer 4) Low Grad Codes	1) ATP-01 Vol 1 2) ATP-31 3) APP-1 4) APP-7 5) AXP-02 6) CFCD-106 7) CFSAS Basic Electronic Warfare Manual

Notes to Instructor

- Voice procedures, block reports and codewords/prowords utilization mistakes shall be corrected throughout.

EO 003.05 – Assess AAW Threat Information

003.05.01 – ZIPPO Criteria

Conditions		Sequence
Classified Classroom		After EO 003.04
Time	**Classification**	**Instructor**
400 minutes	SECRET	NES Op PO1/PO2

Learning Objectives	Method
1) Apply ZIPPO criteria 2) Categorize missile groups 3) Conduct AAW duties	• Self-Study • Guided Discussion

Activity Steps	
	<u>LO 1 to 3: Self-Study & Guided Discussion (400 minutes)</u> 1) Trainees will conduct readings relating to AWW general tactical procedures. 2) Following completion of reading, the instructor will conduct a guided discussion on the following subjects: a. ZIPPO Criteria b. Missile Grouping c. AAW Duties to include EWC

Resources	References
1) Secret standalone workstation 2) NTDC(A) PO 003 Required Reading List 3) White board	1) ATP-01 Vol 1 2) ATP-31 3) ATP-55 4) CFCD-106

Notes to Instructor
• Instructor shall stimulate discussion by asking open format questions relating to the EWS role and responsibilities directly related to the subjects listed above.

003.05.02 – ASMD

Conditions		Sequence
Classified Classroom		After EO 003.05.01
Time	**Classification**	**Instructor**
1400 minutes	SECRET	NES Op PO1/PO2

Learning Objectives	Method
1) Coordinate ASMD reactions 2) Plan ASMD reactions 3) Execute ASMD reactions 4) Apply ASMD maneuvering 5) Conduct Detect to Engage (DTE) sequence; 6) Recognizing FBA profiles; 7) Employ SOFTKILL; 8) Maintain Recognized Air Picture (RAP); 9) Utilize Identification Criteria (ID CRIT) and check prints	• Self-Study • Guided Discussion • Group Assignment

Activity Steps	
	<u>LO 1 to 8: Self-Study & Guided Discussion (400 minutes)</u> 1) Trainees will conduct preliminary readings from the NTDC(A) PO 003 Required Reading List on: a. ASMD b. DTE c. SOFT KILL employment d. RAP 2) Following completion of reading, the instructor will conduct a guided discussion on the following subjects: a. AWW battlespace b. DTE sequence c. ASMD procedures d. FU/FT/FI e. FBA profiles f. SOFT KILL employment g. RAP compilation process <u>LO 9: Group Assignment (1000 minutes)</u> 3) Instructor will divide the class into syndicate groups. Each group will be assigned AWW doctrines proportionate to group size (ex: group of 3 trainees would be assigned 3 AWW doctrines). 4) Each group is to develop a scenario/white setting that would accurately demonstrate each AWW doctrine assigned. When building scenarios/white settings trainees are to ensure that all elements used (platforms, emitters, threat) are taken from the NTDC (A) EW Mission Phase chosen game.

	5) Each group will present their assigned doctrines in an interactive white board presentation. Presentation must take into account the following based on individual requirements; a. ID CRIT and check prints b. POP UP/FU/FT/FI c. Missile group d. ZIPPO criteria e. All relevant tactical voice procedures f. All relevant low grade codes and code words g. Appropriate soft kill employment h. Appropriate ASMD reactions i. RAP considerations

Resources	References
1) Secret standalone workstation 2) NTDC(A) PO 003 Required Reading List 3) White board	1) ATP-01 Vol 1 2) ATP-31 3) ATP-55 4) CFCD-106

Notes to Instructor

- The same AWW doctrine can be assigned to different groups if required.

EO 003.06 – Interpret ASuW General Tactical Procedures

003.06.01 – Interpret ASuW General Tactical Procedures

Conditions		Sequence
Classified Classroom		After EO 003.05
Time	**Classification**	**Instructor**
400 minutes	SECRET	NES Op PO1/PO2

Learning Objectives	Method
1) Interpret AsuW duties 2) Apply promulgated tactical information as it applies to internal and external organizations (C00040) 3) Maintain Recognized Maritime Picture (RMP) (F00010) 4) Utilize Recognition Confidence Level (RCL) 5) Perform Area of Interest (AOI) monitoring (C00450) 6) Amplify track Information (F00010) 7) Track high Interest Vessels (C00660)	• Self-Study • Guided Discussion

Activity Steps	
	<u>LO 1 to 7: Self-Study & Guided Discussion (400 minutes)</u> 1) Trainees will conduct readings relating to ASuW general tactical procedures. 2) Following completion of reading, the instructor will conduct a guided discussion on the following subjects: a. ASuW duties and responsibilities b. Anti-Surface Warfare planning considerations c. RMP compilation process d. COI/CCOI monitoring

Resources	References
1) Secret standalone workstation 2) NTDC(A) PO 003 Required Reading List 3) White board	1) ATP-01 Vol 1 2) ATP-31 3) CFCD-106

Notes to Instructor
• Focus should be put on the detection, localization, recognition, identification and dissemination of friendly and enemy surface contacts.

EO 003.07 – Apply Naval Gun Fire Support Procedures

003.07.01 – Apply Naval Gun Fire Support Procedures

Conditions		Sequence
Classroom/CSNI Lab		After EO 003.05
Time	**Classification**	**Instructor**
200 minutes	SECRET	NES Op PO1/PO2

Learning Objectives	Method
1) Interpret spotting procedures 2) Interpret range safety orders 3) Manipulate Naval Gun Fire software	• Self-Study • Guided Discussion • Interactive Lecture

Activity Steps	
	<u>LO 1 & 2: Self-Study & Guided Discussion (150 minutes)</u> 1) Trainees will conduct readings on Naval Gun Fire Support. 2) Following completion of reading, the instructor will conduct a guided discussion on the following subjects: a. Spotting procedures b. Duties of Bombardment Coordinator (BC) c. Range safety procedures d. Naval Gun Fire software <u>LO 3: EDI (50 minutes)</u> 3) Using the CSNI projector and NTDC(A) Naval Gun Fire Support Software Overview checklist, the instructor will conduct a summary with all trainees and provide a brief explanation on each menu and sub-menu and functions within the NGFS software.

Resources	References
1) Secret standalone workstation 2) NTDC(A) PO 003 Required Reading List 3) NTDC(A) Naval Gun Fire Support Software 4) Overview Check List 5) CNSI lab 6) White board	1) AXP-02 2) ATP-31 3) CFCD-106 4) NCPM 1000

Notes to Instructor
Nil.

EC 003 – AWW Tactics

Scope	Conditions	Sequence
EO 003.01 to 003.07	Classified Classroom	After EO 003.07
Time	**Classification**	**Instructor**
1200 minutes	SECRET	NES Op PO1/PO2

Assessment Steps	
	Input: Trainees will be provided with a copy of EC 003 checklist, be given time to review and ask clarifying questions as required. The instructor will assign an AWW doctrine to each trainee and create a 25 question examination from the NTDC(A) Tactical Examination Question Bank and administer to trainees. **Performance:** Trainees will individually: 1) Develop a scenario/white setting in which they accurately demonstrate the assigned doctrine. When developing their scenario/white setting, trainees shall only use elements from the NTDC(A) PO 004 chosen game. 2) Present their scenario/white setting to the class in an interactive white board presentation. 3) Be assessed on their comprehension of AWW tactics and procedures through a written examination. **Output:** Trainee must achieve 12/15 SATs on non-critical factors and 4/4 critical factors to pass this EC. **Feedback:** Instructor will debrief trainees individually following completion of the EC.

Resources	References
1) EC 003 Check List 2) NTDC(A) PO 003 Required Reading List 3) NTDC(A) Tactic Examination Question Bank 4) Secret standalone workstation 5) White board	Nil.

Notes to Instructor
• It is paramount that the doctrines assigned have not been presented previously during EO 003.06. • This EC is a CRITICAL EC.

Limitations	Nil.
Impact of Failure	IAW NPTORD 4500-54.

Item	EC 003 AWW TACTICS	SAT	UNSAT	Remarks
1. Interactive White Board Presentation				
Doctrine demonstrated accurately		☐	☐	CRITICAL
Correct ZIPPO criteria		☐	☐	CRITICAL
Correct SOFTKILL employment		☐	☐	CRITICAL
Identification criteria		☐	☐	
Check print		☐	☐	
Tactical voice/block reports		☐	☐	
Reports made to appropriate Warfare Commanders		☐	☐	
Duty assignment		☐	☐	
ESM management		☐	☐	
Jamming procedures		☐	☐	
EP coordination		☐	☐	
ES coordination		☐	☐	
EW sub team taskings		☐	☐	
Stimulated classroom discussions		☐	☐	
Tactical Examination		☐	☐	CRITICAL

Score: /15
/4

Trainee must achieve 12/15 SATs on non-critical factors and 4/4 critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the trainee.	**Trainee** I have been debriefed on the result of this assessment.

NTDC(A) Approval – 26/01/2021
NPTG HQ Approval – 27/06/2019
Version No: 3.0
MITE Course ID: 121756

Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

PROTECTED B (when complete)

NTDC(A) Approval – 26/01/2021
NPTG HQ Approval – 27/06/2019
Version No: 3.0
MITE Course ID: 121756

PC 003 – Execute AWW Tactics for EW Operations

Scope	Conditions	Sequence
PO 003	SEACOT/SEAWOLF or Ship	To be assessed on completion of EC 003
Time	**Classification**	**Instructor**
12000	SECRET	NES Op PO1/PO2

Assessment Steps	PC 003 will be a training/developmental/practical assessment based on content from PO 003. PC 003 shall be composed of the following three phases: Phase 1: Training Phase (4080 Minutes) Phase 2: Developmental Run Phase (3960 Minutes 330 Minutes per trainee) Phase 3: Assessment Run Phase (3960 Minutes 330 Minutes per trainee) Phase 1: Training Phase - Will consist of 4080 minutes of fixed time, structured free play. Instructor should begin the Training Phase by slowly presenting trainees with each element they will encounter during phases 2 & 3, increasing in difficulty and complexity as the training progresses, and having as many trainees practicing as time and resources permits. This phase will consist of both instructor mentoring and peer to peer mentoring. NOTE: During the training phase each element shall be presented individually, gradually increasing in complexity until the required proficiency is achieved. Phase 2: Developmental Run Phase – Will consist of 330 minutes per trainee of fixed time which shall consist of 20 minutes for system setup and a 90 minute run profile. Trainees shall be debriefed following each run. Phase 3: Assessment Run Phase – Will consist of 330 minutes per trainee of fixed time which shall consist of 20 minutes for system setup and a 90 minute run profile. Trainees shall be debriefed following each run. During phases 2 & 3, trainees shall be exposed to each of the following on every run: 1. A complex RMP (air and surface) 2. Numerous ESM detections of threat emitter 3. A minimum of 3 occasions where a Criss Cross is ordered/requested and then executed 4. A minimum of 1 and a maximum of 3 subsonic missile raid with weapon release

	5. A minimum of 1 and a maximum of 3 supersonic missile raid with weapon release 6. At least 1 probe or raid 7. One instance where FU/FT/FI is fouled Each trainee will experience each DIMS tasking once during both developmental and assessment phases. The DIMS tasking will promulgate the screen, MLA, and starting position for each run. Once the 90 minutes has expired the run will be complete and scored. **Output**: Trainee must achieve 47/58 SATs overall, SAT on all CRITICAL factors, and pass 2/3 assessment runs. **Feedback**: Positive/proactive analysis, professionalism, communication, leadership, QA, organization, attitude, threat knowledge

Resources	References
1) Nil	1) Game pack OP messages

Notes to Instructor

- Training support shall be IAW NWTP Vol 2.
- Each run, during the developmental and assessment phases will last for 90 minutes during which the trainees will execute the AWW mission. All OPTASKs, OPGENs, OPORDs, Supps, ORBAT and DIMS messages for the scenario are in effect as the governing orders and rules for the operational scenario. DIMS are to be drafted and distributed by instructor a minimum of 1 day prior to presentation to trainees. All other messages to be IAW Campus Atlantic chosen game.
- If items on the checklist are not applicable then the trainee will receive a SAT.
- Execution of the developmental and assessment phase scenarios shall be reactive to trainee courses of action and plans. Instructional instructor and support instructor must be intimately familiar with the scenario documents, and those produced by the trainees for execution of the scenario, to be able to react to the trainees in a plausible and realistic fashion.
- Students should be permitted full access to TAC Pacs or required block reports during this phase.

Limitations	PC 003 will be conducted as an individual assessment.
Impact of Failure	IAW NPTORD 4500-54.

Item	PC 003 Execute AWW Tactics for EW Operations	SAT	UNSAT	Remarks
1. CMS Configuration				
MASS/DLF		☐	☐	
RAMSES		☐	☐	
SIRIUS		☐	☐	
TDAs		☐	☐	
EW filters		☐	☐	
2. Communications				
Brief expectations to EW team		☐	☐	
Monitor pertinent networks		☐	☐	
Monitor chat		☐	☐	
Follow directions given by SWC		☐	☐	**CRITICAL (if failure to do so results in mission failure)**
Set expectations (e.g. SITREPs, priorities)		☐	☐	
Brief ORS		☐	☐	
3. Warfare Commanders (WC)				
Pass all pertinent RSP/RAP information to/from applicable WC		☐	☐	
Brief or confirm SWC on WC directions		☐	☐	
4. Picture Compilation				
Ensure RMP current and accurate		☐	☐	
Ensure RAP current and accurate		☐	☐	
ELISRA/MASS SUP/CIO performance		☐	☐	
Employ RCL		☐	☐	
Employ IDCRIT		☐	☐	
5. Electronic Warfare				
Produce recommendations for action (pre-plan ASMD)				
Employ EW/ESM information to supplement RSP/RAP (e.g. cross fix)		☐	☐	
Deliver voice reports to Warfare Commander		☐	☐	
Provide ESM block report to SWC in timely matter		☐	☐	
Manipulate CMS:				

Item	PC 003 Execute AWW Tactics for EW Operations	SAT	UNSAT	Remarks
Input ESM bearing lines		☐	☐	
Input jamming bearing lines		☐	☐	
Input communications intercept bearing lines		☐	☐	
Input ESM cross fix		☐	☐	
Input communications intercept cross fix		☐	☐	
Update parametric data		☐	☐	
6. ASMD – No Weapon Release (if applicable)				
Make ZIPPO call (trainee may correct call if battle space allows)		☐	☐	
Confirm ASMD course is viable		☐	☐	
Select and report SOFTKILL threat type		☐	☐	
QA MTR		☐	☐	
Assess FU/FT/FI		☐	☐	
RAMSES recommendation		☐	☐	
MASS recommendation		☐	☐	
SOPD recommendation		☐	☐	
7. ASMD – Supersonic Missile Raid (only assessed on weapons release)				
Raid:		1	2	3
Time:				
Missile Name:				

	S	U	S	U	S	U	
Make ZIPPO call (trainee may correct call if battle space allows)	☐	☐	☐	☐	☐	☐	CRITICAL
Confirm ASMD recommendation is viable	☐	☐	☐	☐	☐	☐	CRITICAL (if failure to do so results in mission failure)
Select and report SOFTKILL threat type	☐	☐	☐	☐	☐	☐	
QA MTR	☐	☐	☐	☐	☐	☐	
Assess FU/FT/FI	☐	☐	☐	☐	☐	☐	
RAMSES recommendation	☐	☐	☐	☐	☐	☐	
MASS recommendation	☐	☐	☐	☐	☐	☐	

Item	PC 003 Execute AWW Tactics for EW Operations	SAT			UNSAT			Remarks
SOPD recommendation		☐	☐	☐	☐	☐	☐	
MASS employment		☐	☐	☐	☐	☐	☐	CRITICAL
RAMSES employment		☐	☐	☐	☐	☐	☐	CRITICAL
SOPD employment		☐	☐	☐	☐	☐	☐	CRITICAL
8. ASMD – Subsonic Missile Raid (only assessed on weapons release)								
	Raid:	1		2		3		
	Time:							
	Missile Name:							
		S	U	S	U	S	U	
Make ZIPPO call (trainee may correct call if battle space allows)		☐	☐	☐	☐	☐	☐	CRITICAL
Confirm ASMD recommendation is viable		☐	☐	☐	☐	☐	☐	CRITICAL (if failure to do so results in mission failure)
Select and report SOFTKILL threat type		☐	☐	☐	☐	☐	☐	
QA MTR		☐	☐	☐	☐	☐	☐	
Assess FU/FT/FI		☐	☐	☐	☐	☐	☐	
RAMSES recommendation		☐	☐	☐	☐	☐	☐	
MASS recommendation		☐	☐	☐	☐	☐	☐	
SOPD recommendation		☐	☐	☐	☐	☐	☐	
MASS employment		☐	☐	☐	☐	☐	☐	CRITICAL
RAMSES employment		☐	☐	☐	☐	☐	☐	CRITICAL
SOPD employment		☐	☐	☐	☐	☐	☐	CRITICAL
Score: /58 Trainee must achieve 47/58 SATs overall, SAT on all CRITICAL factors, and pass 2/3 assessment runs.								

Item	PC 003 Execute AWW Tactics for EW Operations	SAT	UNSAT	Remarks

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the trainee.	**Trainee** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

PROTECTED B (when complete)

CHAPTER 3: ASSESSMENT OF TRAINEE

GENERAL

1. All Performance Checks (PCs) are based on POs in Chapter 2 of this QSP. Trainees must achieve the standard to be awarded the qualification. PCs will be administered in the form of practical or written assessments and the trainee must pass all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. Enabling Checks (ECs) are formative assessments that help the trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning and increase the retention of material for longer periods in preparation for the PC. EC results do not factor into a trainee's successful completion of the PO.

3. PCs are summative assessments used to determine if the trainee has met the standard under the conditions (where possible) of the PO. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

4. Assessment of trainees is to be completed IAW the standards delineated in this QSP and IAW CFITES Volume 7 (Evaluation).

PROGRESS MONITORING

5. Monitoring learning is the responsibility of all instructional staff and supervisors. Monitoring, including unsatisfactory progress, shall adhere to CFITES Volume 6 Conduct of Instruction Programmes.

6. Continuous monitoring of trainee progress is required to provide the following:
   a. Early warning of trainee difficulties;
   b. Feedback on the effectiveness of training;
   c. Information for the Training Review Board (TRB) (if required); and
   d. Information to assist in the completion of the CF 377 Course Report.

7. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW Training Establishment (TE) SOs or SOPs.

8. PC failures and supplemental PC failures are to be handled IAW TE SOs and SOPs. Supplemental ECs and PCs shall not be administered within 24 hours of the trainee being informed of their EC or PC failure. A re-test shall be scheduled, at minimum, 24 hours after the trainee is informed of the EC or PC failure.

TRAINING REVIEW BOARD

9. The aim of the TRB process is to review a trainee's suitability for continued training and education, while ensuring that the trainee is afforded a fair and just assessment of all relevant

information before a decision is made. TRBs shall be convened and conducted IAW NPTORD 4500-54 Training Review Board Process.

COURSE REPORTS

10. Trainees shall be graded pass or fail IAW NPTORD 4500-51.

11. Course reports (CF 377) shall be prepared and distributed IAW NPTORD 4500-51. If applicable, training limitations and/or incomplete training must be included in the course report.

12. Narratives shall indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager.

13. A record of trainee counselling is to be maintained by the TE.

References

List of all references used within this QSP

A = CF or other official Canadian government publication

B = Official Allied Force or Allied government publication

C = Commercial publication

D = Other publications

Ref	Short Title	Long Title
A1	AIT Leaner Manual	Alternate Instructional Techniques Leaner Manual
A2	CFCD-114	Royal Canadian Navy Electronic Warfare Orders
A3	CFSAL Military Writing Guide	CFSAL Military Writing Guide
A4	CFCD-106	Maritime Tactical Instructions
A5	CFCD-102	Maritime Command Combat Readiness Requirements
A6	Sea Training Guide	Sea Training Guide
A7	NCPM 1000	Halifax Class Modernization Above Water Warfare Procedures Manual
A8	CFSAS Basic Electronic Warfare Manual	CFSAS Basic Electronic Warfare Manual
B1	APP-11	NATO Message Catalogue
B2	ATP-01 Vol 1	Allied Maritime Tactical Instructions and Procedures
B3	ATP-31	NATO Above Water Warfare Manual
B4	AXP-2	The Allied Maritime Above Water Warfare Exercise Manual
B5	APP-7	Joint Brevity Words Publication

B6	ATP-55	Secret Allied Maritime Tactical Instruction and Procedures
B7	APP-1	Allied Maritime Voice Reporting Procedures
C1	U MAN-1120 SIRIUS manual	U MAN-1120 SIRIUS manual
C2	TRASE2430EW Operational and Technical Manual Draft	TRASE2430EW Operational and Technical Manual Draft
C3	SHINCOM handbook	SHINCOM handbook
C4	Communication ESM handbook	Communication ESM handbook

Task List

Task #	Task Statement	Trg / No Trg	PO
A00010	Authenticate communication transmissions	T	003
A00020	Perform communication checks	T	003
A00030	Configure communication terminal for operation	T	002
A00040	Extract information from Communication Plans (COMPLANS), Emission Control (EMCON) Plans and Frequency Plans	T	001
A00050	Extract information from Communications Plans (COMPLANS), Emissions Control (EMCON) Plans and Frequency Plans	NT	-
A00060	Communicate using tactical voice procedures	T	003
B00010	Classify radar contacts	T	003
B00020	Coordinate the Display of Intelligence Information	T	002
B00070	Perform Intelligence gathering	T	003
B00080	Operate combat systems (CS) equipment	T	002
B00090	Evaluate intelligence information	T	001
B00100	Detect radar contacts	T	003
B00120	Adhere to EMCON restrictions	T	001
B00170	Perform pre-sail checks on equipment	T	002
B00180	Prepare EMCON plans	T	001
B00190	Prepare EMCON Table	T	001
B00200	Prepare operational training exercises	T	001
C00020	Conduct Electronic Attacks	T	003
C00030	Operate EW equipment	T	003
C00040	Apply promulgated tactical information as it applies to internal and external organizations	T	003
C00050	Assess jamming	T	003
C00060	Associate electronic intercept information	T	003
C00070	Carryout Electronic Protect (EP) measures	T	003
C00100	Configure Electro Optic Detection System	T	002
C00150	Configure Infrared Detection System	T	002
C00200	Conduct Electronic Attacks (EA)	NT	-
C00280	Conduct EW operational training	T	001

C00300	Coordinate local EW operations	T	001
C00370	Perform hull to emitter correlation (HULTEC)	T	003
C00400	Disseminate ES information	T	003
C00420	Operate Electro Optic Detection System	T	003
C00440	Operate Infrared Detection System	T	002
C00450	Perform Area of Interest (AOI) monitoring	T	003
C00490	Coordinate ship ES, EA and EP operations	T	003
C00520	Direct ship ES, EA and EP operations	T	3
C00530	Interpret Electronic Warfare orders	T	001
C00540	Perform threat analysis	T	001
C00550	Manipulate EW software	NT	-
C00560	Prepare EW brief utilizing presentation application	T	001
C00580	React to changes in the EW environment	T	001
C00590	Execute Overt/Covert ELINT collection operations	NT	-
C00620	Execute Overt/Covert EW collection operations	T	003
C00630	Operate PC based EW software applications	NT	-
C00640	Track electronic intercepts	T	003
C00650	Interpret EMCON policies	T	001
C00660	Optimize EW sensor employment	T	003
C00680	Manage Communication Intercept System Networks	NT	-
C00720	Perform Electronic deception	T	003
D00010	Operate Anti Surface Warfare (ASuW) weapons and sensors	T	002
D00040	Supervise Fire Control (FC) systems operation	NT	-
D00060	Operate AAW Weapons and Sensors	T	002
E00020	Assess Anti-Air Warfare (AWW) threat information	T	003
F00010	Amplify track information	T	003
F00020	Manipulate databases	NT	-
F00080	Conduct/Participate in technical evaluations, tests and trials	NT	-
F00170	Correlate electronic intercept information	T	003
F00180	Identify Maritime, Air, and Land based platforms	T	001
F00190	Maintain Recognized Maritime Picture (RMP)	T	003
F00210	Operate Command, Control, Communications, Computers and Intelligence (C4I) systems	T	002

F00230	Provide information requested by Request For Information (RFI)	NT	
G00010	Assist in individual training events	T	001
G00020	Conduct individual training events	T	001
G00050	Plan / design training exercises	T	001
G00070	Assess individual training performance	T	001
G00090	Conduct team training events	T	001
G00110	Update training scenarios	T	001
G00140	Assess team training performance	T	001
G00180	Assist in team training events	T	001
G00190	Conduct Unit On Job Training (OJT), Ships Development Package (SDP), Pre-Information Package (PIP)	T	001
G00230	Input training scenarios	T	001

Record of Decisions

Dates: 14 to 17 September 2020

Location: S-15 322

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Hunt	SW	PO1	00115	NTDC(A)
McPhee	SA	PO2	00115	NTDC(A)
Sinclair	DGR	PO2	00115	HMCS Fredericton
Fletcher	AJ	PO2	00115	NFS(A)
Morris	KSA	Lt	00211	NTDC(A)

Working Group Objective:

The objective of this working group was to address the concerns identified in the program evaluation dated 23 April 2020.

Specification Task – No Train Decisions:

Task #	Task Statement	Substantiation
C00680	Manage Communication Intercept System Networks	Board decision was to remove this task as the RCN currently is not part of any Communication Intercept System Network.
D00040	Supervise Fire Control (FC) systems operation	Board decision was to remove this task from this specific course as it is taught on the RQ MS course.
F00080	Conduct/Participate in technical evaluations, tests and trials	Board concluded that this task is taught at the RQ AWWD/AWWO level and the EWS follows given directions during evaluations. No training required.
F00230	Provide information requested by Request For Information (RFI)	Board concluded that this task is accomplished by the AWWD from TRINITY and not at the EWS level.
C00550	Manipulate EW Software	This task is trained at a lower rank level
C00620	Execute Overt/Covert EW collection operations	This task is trained at a lower rank level
C00630	Operate PC based EW software applications	This task is trained at a lower rank level
F00020	Manipulate databases	This task is trained at a lower rank level

Discussions / Decisions:

1. The WG reviewed the PE dated 23 April 2020 and identified the recommendations that could be addressed. Several recommendations were selected and prioritized.
2. Train/no train decisions and POs were not changed.
3. Upon commencement of review the WG identified work needed to be done to the PO 003 Required Reading List. Some of the readings were duplicated, and others perhaps outdated or not relevant to the lesson objectives. As CSNI is required to review many of the references identified, PO2 McPhee will revise this list following completion of the WG. He will then update the timings and references in the lesson guidance after his review.
4. The WG started switching delivery of content in EO 003.01 from individual readings to AIT. The group then split the single lesson in EO 003.02 into 3 lessons for delivery and added AIT. After assessing the time required to continue revising the document in this manner for the WG and the time required for follow-on development, it was determined there was neither time nor resources to accomplish such an in-depth review.
5. Tasks that the WG expected to complete were established as follows:
    a. Create a recommended schedule annex to address the CSNI account issue faced for PO 001. Trainees were not getting their accounts set up in time which caused issues completing the content. In the recommended schedule PO 002 will start the course followed by PO 001 to allow time for the CSNI accounts to be created.
    b. Convert document to 3 Chapter QSP.
    c. Clarify PC 003 assessment guidance.
    d. Reduce subjectivity in PC 003 checklist.
    e. Replace the air defence exercise in EC 001.05 with an EWX.
    f. Clarify EO 003.04 lesson guidance.
6. After reviewing PC 003 the WG had lengthy discussion concerning how to interpret the lesson guidance. With the 4 SMEs present, all 4 had differing interpretations. There was much discussion surrounding how NFS(A) had been interpreting the guidance and it was felt that terminology needed to be clarified to convey the intent more clearly. The term developmental was too unclear and was renamed Phase 1 – Training Phase. The developmental runs and assessment runs were titled Phase 2 & 3 respectively to further clarify.
7. Following a general rewrite of the lesson guidance, the WG moved on to addressing PC 003's checklist. Each item within the previous checklist was given its own checkbox, criticals and significants were removed, and the WG removed criteria that doesn't pertain to the PO2 NES Op's job including the criteria previously listed in the aircraft section. The criteria that remained in threat assessment, teamwork & collaboration, AWW team, SWC liaison, and quality control sections were added in where they would appear within the run. Several factors were pulled out for instructors to give feedback on and those are identified in the assessment guidance.
8. The WG identified 8 critical criteria and 3 conditionally critical criteria. The condition being that this mistake caused mission failure. All SMEs were adamant that if the mission failed due to a mistake on these two items, on the part of the trainee, they would not be confident that that individual could do the job.

9. WRT ASMD, a section was added to account for no weapon being fired as instructors were previously using the supersonic missile and subsonic missile portions of the checklist to mark these reactions. There were several discussions surrounding how to control for the amount of raids during each run. Several concerns were identified including:
    a. Trainees "gaming the game" by not performing their jobs properly after being successful on the maximum number of missiles permitted as per the checklist.
    b. Trainees receiving an inordinate number of raids.
    c. Trainees receiving an insufficient number of raids to properly test their knowledge and skills.
10. After discussing implementing only a minimum or a maximum to prevent "gaming the game" it was decided that a range of 1-3 supersonic and 1-3 subsonic missiles was the best solution. A range would be enough to deter "gaming the game" as trainees will not know if they get a second raid following a first, or a third following a second. The WG rationed that this would keep differences between runs fair despite having different scenarios.
11. Following completion of the checklist the WG decided a total score of 81% or 47/58, all criticals, and passing 2/3 runs were fair assessment criterion for PC success.
12. EC 001.01 referred instructors to use trainee created content in EO 001.05, therefore EO 001.05 was rewritten to remove the air defence exercise and replace it with an EWX. Timings were also reallocated to allow for more preparation. EC 001.05 was updated to reflect the changes.
13. In EO 003.04 the EWX 251 was removed as it takes significant time and was not allowing trainees adequate voice procedure practice. Timings were adjusted to accommodate a brief review of learning objectives and more time for voice procedures.

Follow Up Action:

Item	Action	OPI	Expected Date of Completion
1	Update and refine PO 003 Required Reading List to ensure that only pertinent readings are included.	PO2 McPhee	31 October 20
2	Update timings for readings associated with the PO 003 Required Reading List.	PO2 McPhee	31 October 20
3	Recommended schedule annex.	PO2 McPhee	2 October 20
4	Convert document to FireFly 2.2	Lt Morris	2 October 20

Course History:

Working Group: 02 Oct 2017 – 19 Feb 2018

Surname	Initials	Rank	MOSID	Unit
PO1 Lacroix	F	PO1	00115	NTDC(A)
PO1 McDonald	DC	PO1	00115	NTDC(A)
PO2 Kuffner	NWG	PO2	00115	NTDC(A)
PO2 Kelloway	DJ	PO2	00115	NTDC(A)

1. The objective of this working group is twofold:

    a. Update the RQ PO2 QSP to reflect the objectives of the Future Naval Training System Strategy, and the direction of CO NTDC(A) to include:
        i. Expansion of performance based instruction and assessments;
        ii. Incorporation of adult learning methodologies;
        iii. Maximum inclusion of Scenario Based Training; and
        iv. Maximum divestment of PowerPoint and lecture based instruction.
    b. Trial a new QSP format ("Lesson Guidance") with the following objectives:
        i. Shortening the length (pages) of the QSP;
        ii. Shortening the duration of QSP working groups; and
        iii. Maximize QSP interoperability with MCS School Manager;

2. The working group reviewed the old QSP. The consensus was that several changes were required to be made in order to modernized the course and aligned its content with Electronic Warfare Supervisor's role and responsibilities post FELEX. Based on this fact, the working group came to the decision that the major milestones of the course should be as follows;
    a) Prepare EW correspondence
    b) Operate Multi-Function Workstation (MFW) and EW Sub-Systems
    c) Execute AWW tactics for EW operations
    d) Participate in a tactical examination
    e) Participate in EW mission phase

3. The board discussed the EW sub-systems and the fact that the old QSP did not include DLF employment and still referred to the SHIELD system when discussing softkill and CCS instead of CMS 330. The working group concluded that making these changes would be a high priority as the EWS are currently utilizing these systems in their day to day operations.

4. The board decided that the current practice of instructing this course using visual aids such as PowerPoint is inefficient and that the students will spend most of this portion in a synthetic trainer.

5. The working group discussed at length the fact that EW spectrum knowledge has to be reinforced and to that effect decided to add EO 003.01/02/03 which will focus on EW fundamentals, performing electronic deceptions and coordinating ES/EA and EP operations.

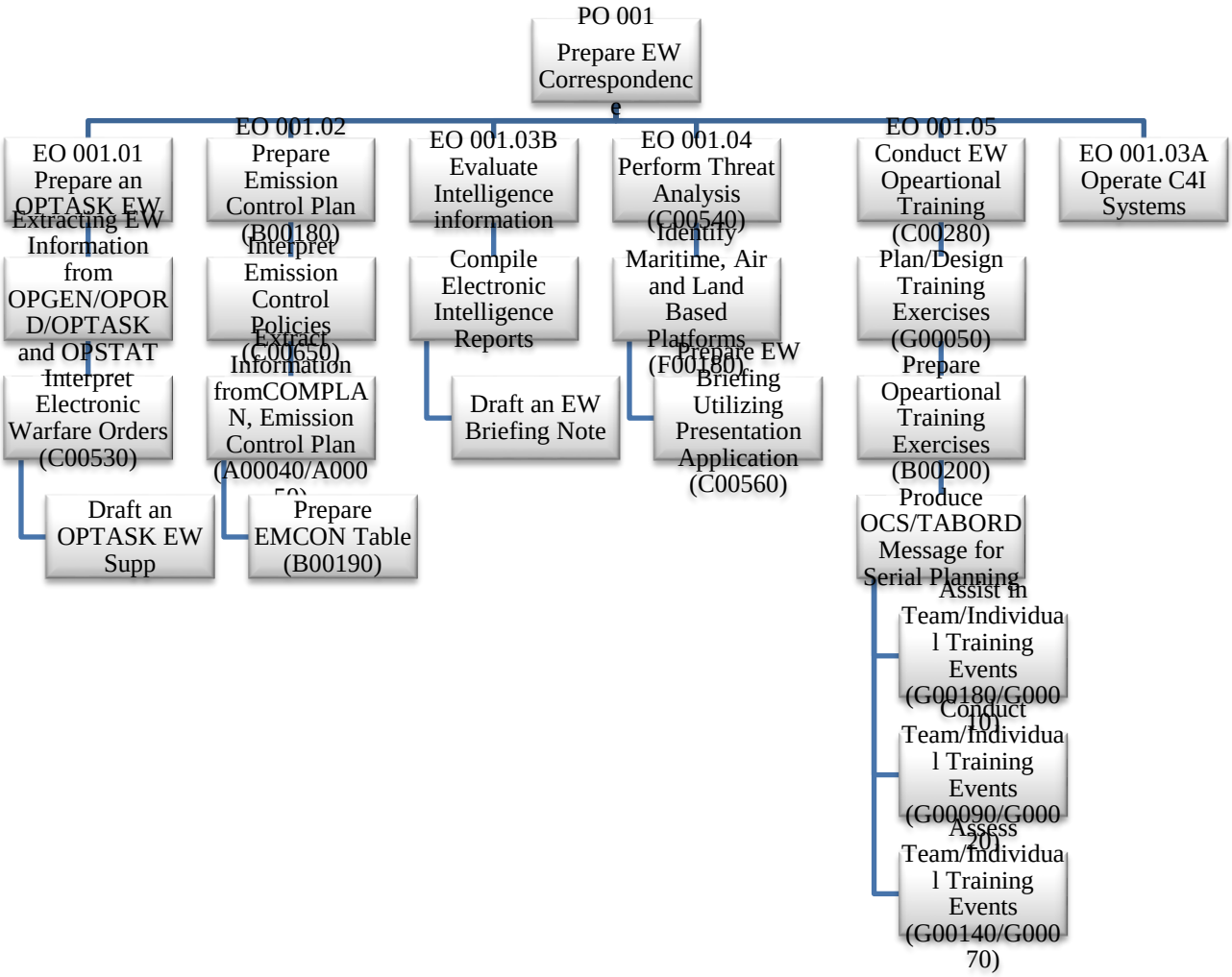
6. When discussing the AWW tactical knowledge, the working group concluded that this portion should be taught using guided discussions followed by syndicate group taskings during which each group would create a scenario demonstrating an assigned RCN doctrine. Then this scenario will be presented to the class with interactive white board presentations.

This methodology will ensure that tactical concept are fully understood by the trainees and applied during interactive white board presentations.

7. The working group concluded that the final PC would constitute of an EW mission phase focusing on EW critical thinking. This phase is modelled on the current RQ AWWD/AWWO course final phase as it was successful and produced strong AWWD and AWWO for the fleet.

8. It was decided to use the task list from JBS and made Train/No train decisions which are documented in the table above.

QSP Scalars



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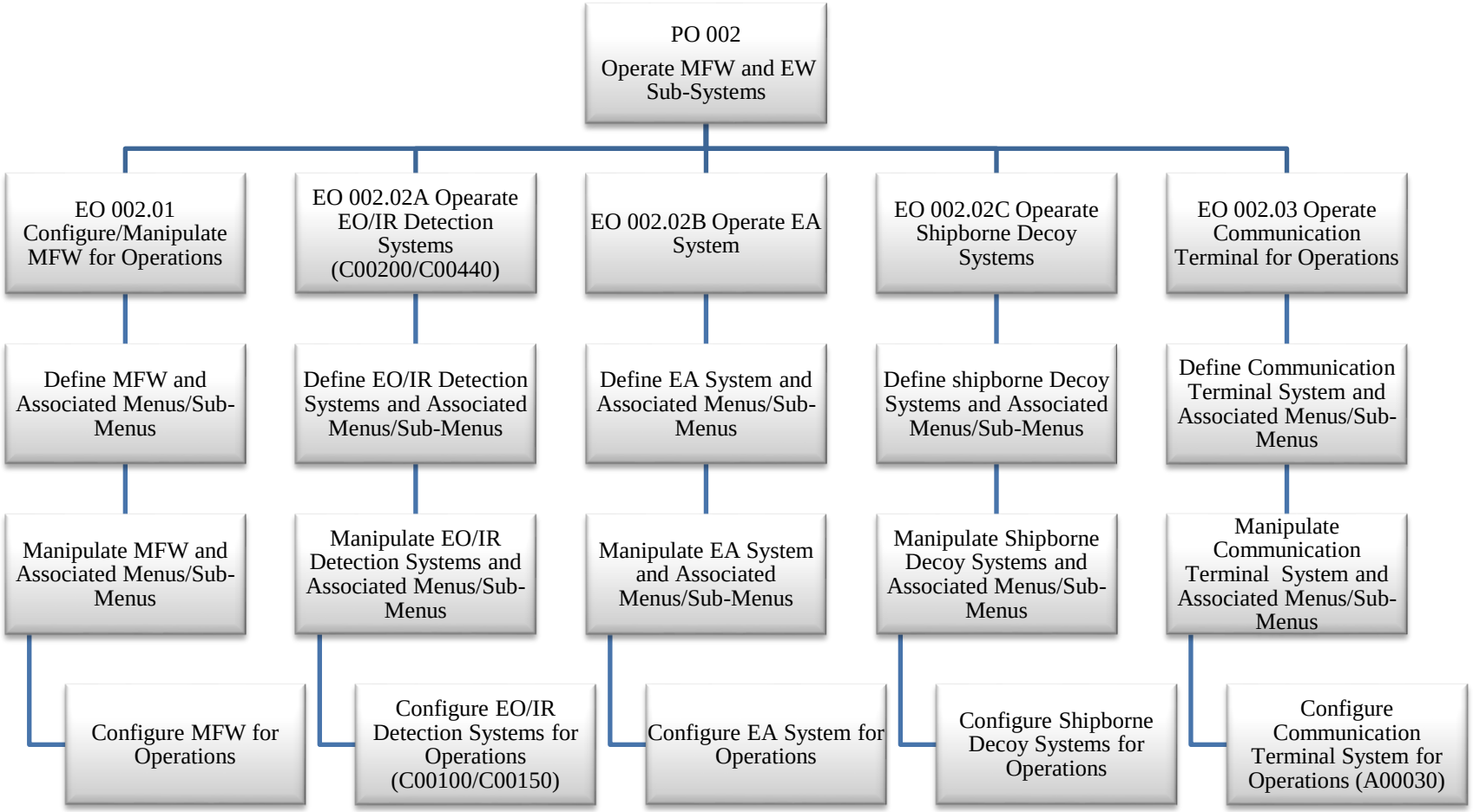

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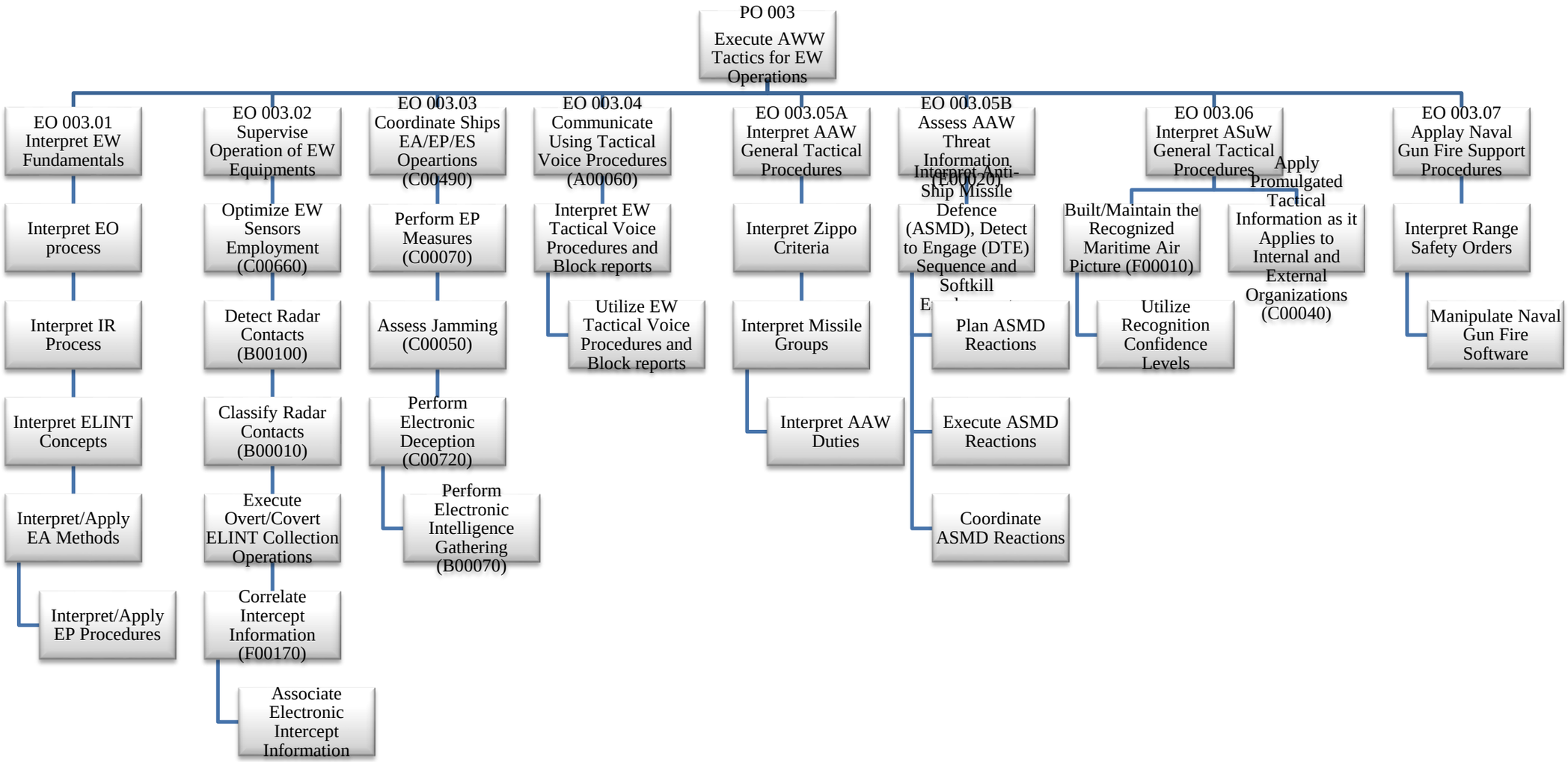

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Costing Information

Course:	Serial:					
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 year total
Airfare (Return)	0	$1,600.00		$0	$0	$0
R&Q	0	$78.75	120	$0	$0	$0
Incidentals	0	$17.30	120	$0	$0	$0
Subtotal				**$0**	**$0**	**$0**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	10	$50.00	1	$500	$500	$5,000.00
Classroom (square meters)	30	$13.00	120	$1560	$1560	$7,800
Computer requirements	12	$0.00	1	$0	$0	$0.00
IT Equip (Proj, Smart board etc)	3	$0.00	1	$0	$0	$0.00
Instructional Materials (Printed / Handouts etc)	8	$30.00	1	$240	$240	$2,400.00
Subtotal				**$2,300**	**$2,300**	**$15,200**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1) (incremental)	0	$0.00	0	$0	$0	$0.00

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Instructor (Lt(N)) (incremental)	0	$0.00	0	$0	$0	$0.00
Incremental Civilian Instructor	0	0	0	$0	$0	$0.00
Subtotal				**$0.00**	**$0.00**	**$0.00**
Total Estimated Costs				**$2,300**	**$2,300**	**$15,200**

Assessment Plan

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria
EC 001.01	EO 001.01	To assess student's preparation of an OPTASK SUPP	Practical	17/22 SATs IAW EC 001.01 Checklist
EC 001.02	EO 001.02	To assess student's understanding of EMCON and creation EMCON tables	Practical	Section I – 8/10 SATs Section II – 1/1 SAT Section III – 3/4 SATs Section IV – 1/1 SAT All section passed IAW EC 001.02 Checklist
EC 001.03	EO 001.03	To assess the student's ability to evaluate intelligence information and draft an EW Briefing Note	Practical	4/5 SATs IAW EC 001.03 Checklist
EC 001.04	EO 001.04	To assess the student's assessment and compilation of threat information and oral presentation skills	Practical	5/7 SATs IAW EC 001.04 Checklist
EC 001.05	EO 001.05	To assess the student's ability to create EW exercises, draft associated messages and execute EW serials	Practical	Section I – 12/15 SATs Section II – 5/6 SATs All section passed IAW EC 001.05 Checklist
PC 001	PO 001	To assess the student's ability to Draft EW related messages, analyse threat information and emission control policies	Practical	Section I – 17/22 SATs Section 2.1 – 8/10 SATs Section 2.2 – 1/1 SAT Section 2.3 – 3/4 SATs Section 2.4 – 1/1 SAT All section passed IAW PC 001 Checklist
EC 002.01	EO 002.01	To assess the student's understanding of CMS 330 and configuring their system for EW operations	Practical	4/5 SATs IAW EC 002.01 Checklist
EC 002.03	EO 002.02 EO 002.03	To assess the student's configuration and manipulation the EO/IR, EA, decoy and communication systems	Practical	Section I – 8/10 SATs Section II – 4/5 SATs Section III – 5/6 SATs

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				Section IV 2/2 SATs Section V – 5/5 SATs Section VI – 5/5 SATs All section passed IAW EC 002.05 Checklist
PC 002	PO 002	To assess the student's understanding of CMS 330 and how to configure their EW sensors and sub-systems for EW operations	Practical	Section I – 4/5 SATs Section II – 8/10 SATs Section III – 4/5 SATs Section IV – 5/6 SATs Section V – 2/2 SATs Section VI – 5/5 SATs Section VII – 5/5 SATs All section passed IAW PC 002 Checklist
EC 003	EO 003.01- EO 003.07	To assess the student's understanding of EW fundamentals and their application of AWW tactics during EW operations	Practical	12/15 SATs 4/4 on the critical factors IAW EC 003 Checklist
PC 003	PO 003	To assess the student's understanding of EW fundamentals and their application of AWW tactics during EW operations through a mission phase	Practical	12/15 SATs overall SAT on all critical factors no more than 1 UNSAT on Significant factors The trainee must achieve 5/8 SAT presentations to pass this PC.

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TEL Development Opportunities

Title	Type	Description of Requirement	Lesson	LSR #
EW Sensors	Video	Ted Talk video by BJ Fizard to cover: 1. Allied Joint EW Doctrine 2. Shipboard EW sensor employments 3. The threat and how it effects EW sensor employment 4. Data fusion and manipulation	003.02.01	

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Scheduling Information

The recommended schedule is based on **maximum load**. Scheduling cells are to increase/decrease course length based on number of trainees enrolled on the course.

Scheduling Annex – RQ-NESOP-PO2													
Lesson		Location	IT Assets					Footprint			Instructor		
#	Time	Location	DWAN	CSNI	SSA	Proj	TEL / Critical Resources	Sec Class	WB	Desks	Type	Qual	Sec Cl
EO 002.01	400	Classroom	y	y	12	y		SEC	n	12	MIL	EO 002.01	400
EO 002.01	800	SEACOT	n	n	12	n		SEC	n	12	MIL	EO 002.01	800
EC 002.01	150	SEACOT	n	n	12	n		SEC	n	12	MIL	EC 002.01	150
EO 002.02.01	300	Classroom	y	y	12	y		SEC	n	12	MIL	EO 002.02.01	300
EO 002.02.01	600	SEACOT and/or ship	n	n	12	n		SEC	n	12	MIL	EO 002.02.01	600
EO 002.02.02	300	Classroom	y	y	12	y		SEC	n	12	MIL	EO 002.02.02	300
EO 002.02.02	300	SEACOT and/or ship	n	n	12	n		SEC	n	12	MIL	EO 002.02.02	300
EO 002.02.03	300	Classroom	y	y	12	y		SEC	n	12	MIL	EO 002.02.03	300
EO 002.02.03	500	SEACOT and/or ship	n	n	12	n		SEC	n	12	MIL	EO 002.02.03	500
EO 002.02.03	200	NAD or ship	n	n	12	n		SEC	n	12	MIL	EO 002.02.03	200
EO 002.03	200	Classroom	y	y	12	y		SEC	n	12	MIL	EO 002.03	200
EO 002.03	150	SEACOT	n	n	12	n		SEC	n	12	MIL	EO 002.03	150
EC 002.02	400	SEACOT	n	n	12	n		SEC	n	12	MIL	EC 002.02	400
PC 002	400	SEACOT	n	n	12	n		SEC	n	12	MIL	PC 002	400
EO 001.01	1050	Classified Classroom	y	y	12	n		SEC	n	12	MIL	EO 001.01	1050

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EC 001.01	400	Classified Classroom	y	y	12	n		SEC	n	12	MIL	EC 001.01	400
EO 001.02	600	Classified Classroom	y	y	12	n		SEC	n	12	MIL	EO 001.02	600
EC 001.02	100	Classified Classroom	y	y	12	n		SEC	n	12	MIL	EC 001.02	100
EO 001.03.01	350	CSNI Lab	y	y	12	n		SEC	n	12	MIL	EO 001.03.01	350
EO 001.03.02	500	CSNI Lab	y	y	12	n		SEC	n	12	MIL	EO 001.03.02	500
EC 001.03	200	CSNI Lab	y	y	12	n		SEC	n	12	MIL	EC 001.03	200
EO 001.04	1000	CSNI Lab	y	y	12	y		SEC	n	12	MIL	EO 001.04	1000
EC 001.04	400	CSNI Lab	y	y	12	n		SEC	n	12	MIL	EC 001.04	400
EO 001.05	500	Classroom	y	y	12	n		SEC	n	12	MIL	EO 001.05	500
EO 001.05	1600	SEACOT	n	n	12	n		SEC	n	12	MIL	EO 001.05	1600
EC 001.05	700	Classroom/ SEACOT	y	y	12	n		SEC	n	12	MIL	EC 001.05	700
PC 001	350	Classroom	y	y	12	n		SEC	n	12	MIL	PC 001	350
EO 003.01	300	Classified Classroom	y	y	12	n		SEC	n	12	MIL	RQ PO2	SEC
EO 003.02	400	Classified Classroom	y	y	12	n		SEC	n	12	MIL	RQ PO2	SEC
EO 003.03	350	Classified Classroom	y	y	12	n		SEC	n	12	MIL	RQ PO2	SEC
EO 003.04	500	Classified Classroom	y	y	12	n		SEC	n	12	MIL	RQ PO2	SEC
EO 003.05.01	400	Classified Classroom	y	y	12	n		SEC	y	12	MIL	RQ PO2	SEC
EO 003.05.02	1400	Classified Classroom	y	y	12	n		SEC	y	12	MIL	RQ PO2	SEC
EO 003.06	400	Classified Classroom	y	y	12	n		SEC	y	12	MIL	RQ PO2	SEC
EO 003.07	200	CSNI lab	y	y	12	n		SEC	y	12	MIL	RQ PO2	SEC

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EC 003	1200	Classified Classroom	y	y	12	n		SEC	y	12	MIL	RQ PO2	SEC
PC 003	12000	SEACOT/SEAWOLF/and or ship	n	n	12	n		SEC	n	12	MIL	RQ PO2	SEC

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Recommended Schedule

Time	DAY 1	DAY 2	DAY 3	DAY 4	DAY 5
Period 1 0800-0850	**Expectations/Admin ISSO, IT, Command Intro (BAT)**	**EO 002.01 Introduction to MFW - Classroom**	**EO 002.01 Introduction to MFW - SEACOT**	**EO 002.01 Introduction to MFW - SEACOT**	**002.02.01 – Introduction to EO/IR sensors - Classroom**
Period 2 0850-0940	**Pre Course Admin**, BAT/TAT	**EO 002.01 Introduction to MFW - Classroom**	**EO 002.01 Introduction to MFW - SEACOT**	**EO 002.01 Introduction to MFW - SEACOT**	**002.02.01 – Introduction to EO/IR sensors - Classroom**
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**Vault (HD)**	**EO 002.01 Introduction to MFW - Classroom**	**EO 002.01 Introduction to MFW - SEACOT**	**EO 002.01 Introduction to MFW - SEACOT**	**002.02.01 – Introduction to EO/IR sensors - Classroom**
Period 4 1040-1130	**EO 002.01 Introduction to MFW - Classroom**	**EO 002.01 Introduction to MFW - SEACOT**	**EO 002.01 Introduction to MFW - SEACOT**	**EO 002.01 Introduction to MFW - SEACOT**	**002.02.01 – Introduction to EO/IR sensors - Classroom**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 002.01 Introduction to MFW - Classroom**	**EO 002.01 Introduction to MFW - SEACOT**	**EO 002.01 Introduction to MFW - SEACOT**	**EC 002.01 – Operate Combat System Equipment**	**002.02.01 – Introduction to EO/IR sensors - SEACOT**
Period 6 1320-1410	**EO 002.01 Introduction to MFW - Classroom**	**EO 002.01 Introduction to MFW - SEACOT**	**EO 002.01 Introduction to MFW - SEACOT**	**EC 002.01 – Operate Combat System Equipment**	**002.02.01 – Introduction to EO/IR sensors - SEACOT**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 002.01 Introduction to MFW - Classroom**	**EO 002.01 Introduction to MFW - SEACOT**	**EO 002.01 Introduction to MFW - SEACOT**	**EC 002.01 – Operate Combat System Equipment**	**002.02.01 – Introduction to EO/IR sensors - SEACOT**
Period 8 1510-1600	**PT**	**PT**	**EO 002.01 Introduction to MFW - SEACOT**	**002.02.01 – Introduction to EO/IR sensors - Classroom**	**PT**

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Time	DAY 6	DAY 7	DAY 8	DAY 9	DAY 10
Period 1 0800-0850	**002.02.01 – Introduction to EO/IR sensors - SEACOT**	**002.02.01 – Introduction to EO/IR sensors - SEACOT**	**002.02.02 – Introduction to EA systems - SEACOT**	**PT**	**002.02.03 – Introduction to shipborne decoy systems - SEACOT**
Period 2 0850-0940	**002.02.01 – Introduction to EO/IR sensors - SEACOT**	**002.02.02 – Introduction to EA systems - Classroom**	**002.02.02 – Introduction to EA systems - SEACOT**		**002.02.03 – Introduction to shipborne decoy systems - SEACOT**
0940-0950	**BREAK**		**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**002.02.01 – Introduction to EO/IR sensors - SEACOT**	**002.02.02 – Introduction to EA systems - Classroom**	**002.02.02 – Introduction to EA systems - SEACOT**	**002.02.03 – Introduction to shipborne decoy systems - Classroom**	**002.02.03 – Introduction to shipborne decoy systems - SEACOT**
002.02.01 – Introduction to EO/IR sensors - Classroom 1040-1130	**002.02.01 – Introduction to EO/IR sensors - SEACOT**	**TAT**	**PT**	**002.02.03 – Introduction to shipborne decoy systems - Classroom**	**BAT**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**002.02.01 – Introduction to EO/IR sensors - SEACOT**	**002.02.02 – Introduction to EA systems - Classroom**	**002.02.02 – Introduction to EA systems - SEACOT**	**002.02.03 – Introduction to shipborne decoy systems - Classroom**	**002.02.03 – Introduction to shipborne decoy systems - SEACOT**
Period 6 1320-1410	**002.02.01 – Introduction to EO/IR sensors - SEACOT**	**002.02.02 – Introduction to EA systems - Classroom**	**002.02.02 – Introduction to EA systems - SEACOT**	**002.02.03 – Introduction to shipborne decoy systems - Classroom**	**002.02.03 – Introduction to shipborne decoy systems - SEACOT**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**002.02.01 – Introduction to EO/IR sensors - SEACOT**	**002.02.02 – Introduction to EA systems - Classroom**	**002.02.02 – Introduction to EA systems - SEACOT**	**002.02.03 – Introduction to shipborne decoy systems - Classroom**	**002.02.03 – Introduction to shipborne decoy systems - SEACOT**
Period 8 1510-1600	**002.02.01 – Introduction to EO/IR sensors - SEACOT**	**002.02.02 – Introduction to EA systems - Classroom**	**002.02.03 – Introduction to shipborne decoy systems - Classroom**	**002.02.03 – Introduction to shipborne decoy systems - SEACOT**	**002.02.03 – Introduction to shipborne decoy systems - SEACOT**

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Time	DAY 11	DAY 12	DAY 13	DAY 14	DAY 15
Period 1 0800-0850	**002.02.03 – Introduction to shipborne decoy systems - SEACOT**	**002.03.01 – Introduction to SHINCOM - Classroom**	**EC 002.03– Operate ASuW Systems and Communications**	**PC 002 – Operate Electronic Warfare Systems**	**PC 002 – Operate Electronic Warfare Systems**
Period 2 0850-0940	**002.02.03 – Introduction to shipborne decoy systems - SEACOT**	**002.03.01 – Introduction to SHINCOM - Classroom**	**EC 002.03– Operate ASuW Systems and Communications**	**PC 002 – Operate Electronic Warfare Systems**	**001.01.01 – Operational Messages**
0940-0950	**BREAK**				**BREAK**
Period 3 0950-1040	**PT**	**002.03.01 – Introduction to SHINCOM - Classroom**	**EC 002.03– Operate ASuW Systems and Communications**	**PC 002 – Operate Electronic Warfare Systems**	**001.01.01 – Operational Messages**
Period 4 1040-1130		**002.03.01 – Introduction to SHINCOM - Classroom**	**EC 002.03– Operate ASuW Systems and Communications**	**TAT**	**BAT**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**002.02.03 – Introduction to shipborne decoy systems - Ship/NAD**	**002.03.01 – Introduction to SHINCOM - SEACOT**	**EC 002.03– Operate ASuW Systems and Communications**	**PC 002 – Operate Electronic Warfare Systems**	**001.01.01 – Operational Messages**
Period 6 1320-1410	**002.02.03 – Introduction to shipborne decoy systems - Ship/NAD**	**002.03.01 – Introduction to SHINCOM - SEACOT**	**EC 002.03– Operate ASuW Systems and Communications**	**PC 002 – Operate Electronic Warfare Systems**	**001.01.01 – Operational Messages**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**002.02.03 – Introduction to shipborne decoy systems - Ship/NAD**	**002.03.01 – Introduction to SHINCOM - SEACOT**	**EC 002.03– Operate ASuW Systems and Communications**	**PC 002 – Operate Electronic Warfare Systems**	**001.01.01 – Operational Messages**
Period 8 1510-1600	**002.02.03 – Introduction to shipborne decoy systems - Ship/NAD**	**PT**	**EC 002.03– Operate ASuW Systems and Communications**	**PC 002 – Operate Electronic Warfare Systems**	**001.01.01 – Operational Messages**

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Time	DAY 16	DAY 17	DAY 18	DAY 19	DAY 20
Period 1 0800-0850	**001.01.01 – Operational Messages**	**001.01.01 – Operational Messages**	**EC 001.01 – Prepare OPTASK EW SUPP**	**PT**	**001.02.01 – Electronic Warfare Planning**
Period 2 0850-0940	**001.01.01 – Operational Messages**	**001.01.01 – Operational Messages**	**EC 001.01 – Prepare OPTASK EW SUPP**		**001.02.01 – Electronic Warfare Planning**
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**001.01.01 – Operational Messages**	**001.01.01 – Operational Messages**	**EC 001.01 – Prepare OPTASK EW SUPP**	**EC 001.01 – Prepare OPTASK EW SUPP**	**001.02.01 – Electronic Warfare Planning**
Period 4 1040-1130	**001.01.01 – Operational Messages**	**TAT**	**PT**	**001.02.01 – Electronic Warfare Planning**	**BAT**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**001.01.01 – Operational Messages**	**001.01.01 – Operational Messages**	**EC 001.01 – Prepare OPTASK EW SUPP**	**001.02.01 – Electronic Warfare Planning**	**001.02.01 – Electronic Warfare Planning**
Period 6 1320-1410	**001.01.01 – Operational Messages**	**001.01.01 – Operational Messages**	**EC 001.01 – Prepare OPTASK EW SUPP**	**001.02.01 – Electronic Warfare Planning**	**001.02.01 – Electronic Warfare Planning**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**001.01.01 – Operational Messages**	**001.01.01 – Operational Messages**	**EC 001.01 – Prepare OPTASK EW SUPP**	**001.02.01 – Electronic Warfare Planning**	**001.02.01 – Electronic Warfare Planning**
Period 8 1510-1600	**001.01.01 – Operational Messages**	**001.01.01 – Operational Messages**	**EC 001.01 – Prepare OPTASK EW SUPP**	**001.02.01 – Electronic Warfare Planning**	**001.02.01 – Electronic Warfare Planning**

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Time	DAY 21	DAY 22	DAY 23	DAY 24	DAY 25
Period 1 0800-0850	**EC 001.02 – Prepare EMCON TABLE**	**001.03.01 – Operate C4I Systems**	**001.03.02 – Intelligence Exploitation**	**PT**	**001.04.01 – Order of Battle**
Period 2 0850-0940	**EC 001.02 – Prepare EMCON TABLE**	**001.03.02 – Intelligence Exploitation**	**001.03.02 – Intelligence Exploitation**		**001.04.01 – Order of Battle**
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**001.03.01 – Operate C4I Systems**	**001.03.02 – Intelligence Exploitation**	**001.03.02 – Intelligence Exploitation**	**EC 001.03 – Evaluate Intelligence Information**	**001.04.01 – Order of Battle**
Period 4 1040-1130	**001.03.01 – Operate C4I Systems**	**TAT**	**PT**	**001.04.01 – Order of Battle**	**BAT**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**001.03.01 – Operate C4I Systems**	**001.03.02 – Intelligence Exploitation**	**001.03.02 – Intelligence Exploitation**	**001.04.01 – Order of Battle**	**001.04.01 – Order of Battle**
Period 6 1320-1410	**001.03.01 – Operate C4I Systems**	**001.03.02 – Intelligence Exploitation**	**EC 001.03 – Evaluate Intelligence Information**	**001.04.01 – Order of Battle**	**001.04.01 – Order of Battle**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**001.03.01 – Operate C4I Systems**	**001.03.02 – Intelligence Exploitation**	**EC 001.03 – Evaluate Intelligence Information**	**001.04.01 – Order of Battle**	**001.04.01 – Order of Battle**
Period 8 1510-1600	**001.03.01 – Operate C4I Systems**	**001.03.02 – Intelligence Exploitation**	**EC 001.03 – Evaluate Intelligence Information**	**001.04.01 – Order of Battle**	**001.04.01 – Order of Battle**

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Time	DAY 26	DAY 27	DAY 28	DAY 29	DAY 30
Period 1 0800-0850	**001.04.01 – Order of Battle**	**EC 001.04 – Perform Threat Analysis**	**001.05.01 – Training Initiatives - Classroom**	**PT**	**001.05.01 – Training Initiatives - SEACOT**
Period 2 0850-0940	**001.04.01 – Order of Battle**	**EC 001.04 – Perform Threat Analysis**	**001.05.01 – Training Initiatives - Classroom**		**001.05.01 – Training Initiatives - SEACOT**
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**001.04.01 – Order of Battle**	**EC 001.04 – Perform Threat Analysis**	**001.05.01 – Training Initiatives - Classroom**	**001.05.01 – Training Initiatives - Classroom**	**TAT**
Period 4 1040-1130	**001.04.01 – Order of Battle**	**EC 001.04 – Perform Threat Analysis**	**PT**	**001.05.01 – Training Initiatives - Classroom**	**BAT**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**001.04.01 – Order of Battle**	**EC 001.04 – Perform Threat Analysis**	**001.05.01 – Training Initiatives - Classroom**	**001.05.01 – Training Initiatives - Classroom**	**001.05.01 – Training Initiatives - SEACOT**
Period 6 1320-1410	**001.04.01 – Order of Battle**	**EC 001.04 – Perform Threat Analysis**	**001.05.01 – Training Initiatives - Classroom**	**001.05.01 – Training Initiatives - SEACOT**	**001.05.01 – Training Initiatives - SEACOT**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**001.04.01 – Order of Battle**	**EC 001.04 – Perform Threat Analysis**	**001.05.01 – Training Initiatives - Classroom**	**001.05.01 – Training Initiatives - SEACOT**	**001.05.01 – Training Initiatives - SEACOT**
Period 8 1510-1600	**001.04.01 – Order of Battle**	**EC 001.04 – Perform Threat Analysis**	**001.05.01 – Training Initiatives - Classroom**	**001.05.01 – Training Initiatives - SEACOT**	**001.05.01 – Training Initiatives - SEACOT**

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Time	DAY 31	DAY 32	DAY 33	DAY 34	DAY 35
Period 1 0800-0850	**001.05.01 – Training Initiatives - SEACOT**	**001.05.01 – Training Initiatives - SEACOT**	**001.05.01 – Training Initiatives - SEACOT**	**PT**	**EC 001.05 – Coordinate EW Operational Training**
Period 2 0850-0940	**001.05.01 – Training Initiatives - SEACOT**	**001.05.01 – Training Initiatives - SEACOT**	**001.05.01 – Training Initiatives - SEACOT**		**EC 001.05 – Coordinate EW Operational Training**
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**001.05.01 – Training Initiatives - SEACOT**	**001.05.01 – Training Initiatives - SEACOT**	**001.05.01 – Training Initiatives - SEACOT**	**001.05.01 – Training Initiatives - SEACOT**	**EC 001.05 – Coordinate EW Operational Training**
Period 4 1040-1130	**001.05.01 – Training Initiatives - SEACOT**	**TAT**	**PT**	**EC 001.05 – Coordinate EW Operational Training**	**BAT**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**001.05.01 – Training Initiatives - SEACOT**	**001.05.01 – Training Initiatives - SEACOT**	**001.05.01 – Training Initiatives - SEACOT**	**EC 001.05 – Coordinate EW Operational Training**	**EC 001.05 – Coordinate EW Operational Training**
Period 6 1320-1410	**001.05.01 – Training Initiatives - SEACOT**	**001.05.01 – Training Initiatives - SEACOT**	**001.05.01 – Training Initiatives - SEACOT**	**EC 001.05 – Coordinate EW Operational Training**	**EC 001.05 – Coordinate EW Operational Training**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**001.05.01 – Training Initiatives - SEACOT**	**001.05.01 – Training Initiatives - SEACOT**	**001.05.01 – Training Initiatives - SEACOT**	**EC 001.05 – Coordinate EW Operational Training**	**EC 001.05 – Coordinate EW Operational Training**
Period 8 1510-1600	**001.05.01 – Training Initiatives - SEACOT**	**001.05.01 – Training Initiatives - SEACOT**	**001.05.01 – Training Initiatives - SEACOT**	**EC 001.05 – Coordinate EW Operational Training**	**EC 001.05 – Coordinate EW Operational Training**

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Time	DAY 36	DAY 37	DAY 38	DAY 39	DAY 40
Period 1 0800-0850	**EC 001.05 – Coordinate EW Operational Training**	**PC 001 – Prepare Electronic Warfare Correspondence**	**003.01.01 – Interpret Electronic Warfare Theory**	**PT**	**003.03.01 – Directing Ships EA/EP/ES Operations**
Period 2 0850-0940	**EC 001.05 – Coordinate EW Operational Training**	**003.01.01 – Interpret Electronic Warfare Theory**	**003.02.01 – Applying EW Procedures**		**003.03.01 – Directing Ships EA/EP/ES Operations**
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**PC 001 – Prepare Electronic Warfare Correspondence**	**003.01.01 – Interpret Electronic Warfare Theory**	**003.02.01 – Applying EW Procedures**	**003.02.01 – Applying EW Procedures**	**003.03.01 – Directing Ships EA/EP/ES Operations**
Period 4 1040-1130	**PC 001 – Prepare Electronic Warfare Correspondence**	**TAT**	**PT**	**003.02.01 – Applying EW Procedures**	**BAT**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**PC 001 – Prepare Electronic Warfare Correspondence**	**003.01.01 – Interpret Electronic Warfare Theory**	**003.02.01 – Applying EW Procedures**	**003.03.01 – Directing Ships EA/EP/ES Operations**	**003.04.01 – Communicate Using Tactical Voice Procedures**
Period 6 1320-1410	**PC 001 – Prepare Electronic Warfare Correspondence**	**003.01.01 – Interpret Electronic Warfare Theory**	**003.02.01 – Applying EW Procedures**	**003.03.01 – Directing Ships EA/EP/ES Operations**	**003.04.01 – Communicate Using Tactical Voice Procedures**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**PC 001 – Prepare Electronic Warfare Correspondence**	**003.01.01 – Interpret Electronic Warfare Theory**	**003.02.01 – Applying EW Procedures**	**003.03.01 – Directing Ships EA/EP/ES Operations**	**003.04.01 – Communicate Using Tactical Voice Procedures**
Period 8 1510-1600	**PC 001 – Prepare Electronic Warfare Correspondence**	**003.01.01 – Interpret Electronic Warfare Theory**	**003.02.01 – Applying EW Procedures**	**003.03.01 – Directing Ships EA/EP/ES Operations**	**003.04.01 – Communicate Using Tactical Voice Procedures**

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Time	DAY 41	DAY 42	DAY 43	DAY 44	DAY 45
Period 1 0800-0850	**003.04.01 – Communicate Using Tactical Voice Procedures**	**003.05.01 – ZIPPO Criteria**	**003.05.02 – ASMD**	**PT**	**003.05.02 – ASMD**
Period 2 0850-0940	**003.04.01 – Communicate Using Tactical Voice Procedures**	**003.05.01 – ZIPPO Criteria**	**003.05.02 – ASMD**		**003.05.02 – ASMD**
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**003.04.01 – Communicate Using Tactical Voice Procedures**	**003.05.01 – ZIPPO Criteria**	**003.05.02 – ASMD**	**003.05.02 – ASMD**	**003.05.02 – ASMD**
Period 4 1040-1130	**003.04.01 – Communicate Using Tactical Voice Procedures**	**TAT**	**PT**	**003.05.02 – ASMD**	**BAT**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**003.04.01 – Communicate Using Tactical Voice Procedures**	**003.05.01 – ZIPPO Criteria**	**003.05.02 – ASMD**	**003.05.02 – ASMD**	**003.05.02 – ASMD**
Period 6 1320-1410	**003.04.01 – Communicate Using Tactical Voice Procedures**	**003.05.01 – ZIPPO Criteria**	**003.05.02 – ASMD**	**003.05.02 – ASMD**	**003.05.02 – ASMD**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**003.05.01 – ZIPPO Criteria**	**003.05.01 – ZIPPO Criteria**	**003.05.02 – ASMD**	**003.05.02 – ASMD**	**003.05.02 – ASMD**
Period 8 1510-1600	**003.05.01 – ZIPPO Criteria**	**003.05.02 – ASMD**	**003.05.02 – ASMD**	**003.05.02 – ASMD**	**003.05.02 – ASMD**

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Time	DAY 46	DAY 47	DAY 48	DAY 49	DAY 50
Period 1 0800-0850	**003.05.02 – ASMD**	**003.06.01 – Interpret ASuW General Tactical Procedures**	**003.07.01 – Apply Naval Gun Fire Support Procedures**	**PT**	**EC 003 – AWW Tactics**
Period 2 0850-0940	**003.05.02 – ASMD**	**003.06.01 – Interpret ASuW General Tactical Procedures**	**003.07.01 – Apply Naval Gun Fire Support Procedures**		**EC 003 – AWW Tactics**
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**003.05.02 – ASMD**	**003.06.01 – Interpret ASuW General Tactical Procedures**	**003.07.01 – Apply Naval Gun Fire Support Procedures**	**EC 003 – AWW Tactics**	**EC 003 – AWW Tactics**
Period 4 1040-1130	**003.05.02 – ASMD**	**TAT**	**003.07.01 – Apply Naval Gun Fire Support Procedures**	**EC 003 – AWW Tactics**	**BAT**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**003.05.02 – ASMD**	**003.06.01 – Interpret ASuW General Tactical Procedures**	**PT**	**EC 003 – AWW Tactics**	**EC 003 – AWW Tactics**
Period 6 1320-1410	**003.05.02 – ASMD**	**003.06.01 – Interpret ASuW General Tactical Procedures**	**EC 003 – AWW Tactics**	**EC 003 – AWW Tactics**	**EC 003 – AWW Tactics**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**003.05.02 – ASMD**	**003.06.01 – Interpret ASuW General Tactical Procedures**	**EC 003 – AWW Tactics**	**EC 003 – AWW Tactics**	**EC 003 – AWW Tactics**
Period 8 1510-1600	**003.06.01 – Interpret ASuW General Tactical Procedures**	**003.06.01 – Interpret ASuW General Tactical Procedures**	**EC 003 – AWW Tactics**	**EC 003 – AWW Tactics**	**EC 003 – AWW Tactics**

ANNEX I – RECOMMENDED SCHEDULE

Time	DAY 51	DAY 52	DAY 53	DAY 54	DAY 55
Period 1 0800-0850	**EC 003 – AWW Tactics**	**PT**	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PT**	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase
Period 2 0850-0940	**EC 003 – AWW Tactics**	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase		**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EC 003 – AWW Tactics**	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase
Period 4 1040-1130	**EC 003 – AWW Tactics**	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EC 003 – AWW Tactics**	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase
Period 6 1320-1410	**EC 003 – AWW Tactics**	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EC 003 – AWW Tactics**	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**BAT**
Period 8 1510-1600	**EC 003 – AWW Tactics**	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**TAT**

Time	DAY 56	DAY 57	DAY 58	DAY 59	DAY 60
Period 1 0800-0850	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PT**	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PT**	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase
Period 2 0850-0940	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase		**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase
Period 4 1040-1130	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase
Period 6 1320-1410	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**BAT**
Period 8 1510-1600	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**TAT**

Time	DAY 61	DAY 62	DAY 63	DAY 64	DAY 65
Period 1 0800-0850	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PT**	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PT**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**
Period 2 0850-0940	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase		**PC 003 – Execute AWW Tactics for EW Operations - Developmental**
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**
Period 4 1040-1130	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**
Period 6 1320-1410	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**BAT**
Period 8 1510-1600	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations** - Training Phase	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**TAT**

Time	DAY 66	DAY 67	DAY 68	DAY 69	DAY 70
Period 1 0800-0850	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PT**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PT**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**
Period 2 0850-0940	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**		**PC 003 – Execute AWW Tactics for EW Operations - Developmental**
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**
Period 4 1040-1130	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**
Period 6 1320-1410	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**BAT**
Period 8 1510-1600	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**TAT**

Time	DAY 71	DAY 72	DAY 73	DAY 74	DAY 75
Period 1 0800-0850	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PT**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
Period 2 0850-0940	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**		**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
0940-0950	**BREAK**	BREAK	BREAK	BREAK	BREAK
Period 3 0950-1040	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
Period 4 1040-1130	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**TAT**	**PT**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**BAT**
1130-1230	**LUNCH**	LUNCH	LUNCH	LUNCH	LUNCH
Period 5 1230-1320	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
Period 6 1320-1410	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
1410-1420	**BREAK**	BREAK	BREAK	BREAK	BREAK
Period 7 1420-1510	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
Period 8 1510-1600	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Developmental**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**

Time	DAY 76	DAY 77	DAY 78	DAY 79	DAY 80
Period 1 0800-0850	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PT**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
Period 2 0850-0940	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**		**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
0940-0950	**BREAK**	BREAK	BREAK	BREAK	BREAK
Period 3 0950-1040	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
Period 4 1040-1130	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**TAT**	**PT**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**BAT**
1130-1230	**LUNCH**	LUNCH	LUNCH	LUNCH	LUNCH
Period 5 1230-1320	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
Period 6 1320-1410	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
1410-1420	**BREAK**	BREAK	BREAK	BREAK	BREAK
Period 7 1420-1510	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
Period 8 1510-1600	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**

Time	DAY 81	DAY 82	DAY 83	DAY 84	DAY 85
Period 1 0800-0850	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PT**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
Period 2 0850-0940	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**		**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
0940-0950	**BREAK**	BREAK	BREAK	BREAK	BREAK
Period 3 0950-1040	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
Period 4 1040-1130	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**TAT**	**PT**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**BAT**
1130-1230	**LUNCH**	LUNCH	LUNCH	LUNCH	LUNCH
Period 5 1230-1320	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
Period 6 1320-1410	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
1410-1420	**BREAK**	BREAK	BREAK	BREAK	BREAK
Period 7 1420-1510	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
Period 8 1510-1600	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**

Time	DAY 85
Period 1 0800-0850	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
Period 2 0850-0940	**PC 003 – Execute AWW Tactics for EW Operations - Assessments**
0940-0950	BREAK
Period 3 0950-1040	**Admin/Graduation**
Period 4 1040-1130	
1130-1230	LUNCH
Period 5 1230-1320	**Admin/Graduation**
Period 6 1320-1410	
1410-1420	BREAK
Period 7 1420-1510	
Period 8 1510-1600	